

RESEARCH ARTICLE

Can AI Observe Geographical Space? GVSABench For Evaluating Geo-Visuospatial Ability of Large Multimodal Models

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Correspondence: Hua Liao (liaohua@hunnu.edu.cn)**Received:** 2 October 2025 | **Revised:** 14 December 2025 | **Accepted:** 7 January 2026**Keywords:** benchmark evaluation | geo-visuospatial ability (GVSA) | multimodal large language models (MLLMs) | prompting strategies | spatial reasoning

ABSTRACT

This study introduces GVSABench, a comprehensive Geo-Visuospatial Ability (GVSA) benchmark for evaluating the spatial abilities of multimodal large language models (MLLMs). The benchmark systematically spans intrinsic and extrinsic, static and dynamic, and geographic and non-geographic dimensions, comprising 851 image-based tasks. These tasks cover a variety of tasks including spatial visualization, spatial relation reasoning, scene interpretation, spatial orientation and localization, and map-based problem-solving. Seven state-of-the-art MLLMs were tested under zero-shot, zero-CoT, and one-shot prompting strategies. Results indicate that overall accuracies remain moderate to low, with significant variability across models, languages, and task types. Prompting strategies yield only limited improvements, underscoring that engineering alone cannot compensate for fundamental deficits in spatial cognition. Moreover, a scale-separation effect was observed, with distinct performance patterns between geographic and non-geographic tasks, as well as between small- and large-scale contexts. These findings reveal the incomplete integration of visual, linguistic, and spatial reasoning in current MLLMs. GVSABench offers a reproducible and cognitively grounded framework for advancing future research on robust and human-aligned spatial intelligence.

1 | Introduction

With the rapid development of artificial intelligence (AI), particularly large language models (LLMs) and multimodal LLMs (MLLMs), their impressive performance in natural language processing, data analysis, and code generation has attracted extensive attention from academia and industry (Chang et al. 2024; Xu, Hu, et al. 2025). As these models are increasingly applied in real-world applications, evaluating their capabilities has become a critical priority (Mooney et al. 2023; Hochmair et al. 2024; Xu, Hu, et al. 2025). Among these capabilities, geo-visuospatial ability (GVSA)—the capacity to perceive, interpret, and reason with geographic information represented in visual-spatial formats

such as maps, diagrams, and geospatial imagery—plays a central role.

GVSA can be viewed as a specialized extension of spatial ability. Whereas spatial ability broadly includes cognitive skills such as mental rotation, visualization, and perspective-taking, geo-visuospatial ability emphasizes geographic contexts (Golledge 2002; National Research Council 2006; Bednarz and Lee 2012). It integrates visuospatial processing with geographic knowledge, enabling individuals to interpret cartographic symbols, reason across scales, visualize transformations of geospatial data, and navigate complex environments. Such abilities are essential in geography, geographic information science (GIS),

urban planning, and Earth sciences, where geovisualizations serve as primary media for reasoning and decision-making.

Recent research in GIS has begun to assess the geo-visuospatial capacities of LLMs and MLLMs through two main approaches. The first examines language-based spatial reasoning, including spatial question answering, geographic knowledge extraction, textual spatial relation tasks, and code generation (Mooney et al. 2023; Hu et al. 2023; Gramacki et al. 2024; Yamada et al. 2024; Zhang, Dong, et al. 2024; Yang, Fu, et al. 2025; Xu, Hu, et al. 2025; Ji et al. 2025; Hou et al. 2025). These approaches test how well models can interpret or express geographic concepts through language alone. The second focuses on visual inputs and multimodal tasks, such as chart interpretation, diagram reasoning, and map reading (Tao and Xu 2023; J. Xu and Tao 2024; Li, Yao, et al. 2024; Ranasinghe et al. 2025; Danish et al. 2025; Zhang, Sun, et al. 2025; Xu, Zhao, et al. 2025; Hong et al. 2025). However, these studies often focus on task completion rather than assessing the underlying, multidimensional abilities that humans draw upon in geo-visuospatial reasoning.

In psychology, spatial ability has long been studied as a multidimensional construct (Uttal et al. 2013; Malanchini et al. 2020). Frameworks such as Uttal et al.'s (2013) classification of intrinsic versus extrinsic and static versus dynamic skills offer conceptual foundations for systematic evaluation. Yet their geographic extensions remain underdeveloped, limiting current GIS methods to ad hoc assessments. Bridging psychology and geography is therefore essential for cognitively grounded evaluations of geo-visuospatial ability (Jansen 2009; Bednarz and Lee 2012; Ishikawa 2013; Dong et al. 2023).

To address this gap, we extend visuospatial ability into the geographic domain, defining GVSA as the capacity to (1) extract spatial primitives (location, orientation, distance, shape) from geospatial representations, (2) perform visual transformations such as two- and three-dimensional rotation, scaling, reflection, and perspective change, and (3) interpret and reason with geographically grounded visualizations. Building on this definition, we propose an adapted evaluation framework that extends spatial cognition dimensions (Uttal et al. 2013) to encompass geographic complexity.

Based on this framework, we construct the GVSABench, a benchmark dataset comprising eight task categories and 851 instances, ranging from diagram interpretation, map reading to geospatial navigation. We then systematically evaluate seven state-of-the-art MLLMs, including Hunyuan-Vision, QWEN-VL-Plus, GLM-4V-Plus, Gemini-2.0, Claude-3.5-Sonnet, GPT-4o, and DeepSeek-VL2, under various prompting strategies such as zero-shot, one-shot, and chain-of-thought prompting.

2 | Related Work

2.1 | Spatial Ability and Its Evaluation in the Pre-LLM Era

Understanding spatial ability has long been a central focus in cognitive psychology and geography, and scholars have proposed theories on its components (Golledge 2002; Gersmehl and

Gersmehl 2007; Uttal et al. 2013; Malanchini et al. 2020). One of the earliest classifications, proposed by McGee (1979), divided spatial ability into spatial visualization and spatial orientation. The National Research Council (2006) further expanded this view by identifying spatial thinking as consisting of three interrelated components: spatial concepts, representation tools, and reasoning processes. Compared to this broader notion, spatial ability is often defined more narrowly as involving basic cognitive operations independent of external tools. For example, Buckley et al. (2018) identified 27 types of spatial capabilities encompassing both static and dynamic factors. Similarly, Malanchini et al. (2020) grouped 16 spatial skills into three domains: object manipulation, visualization, and navigation.

A significant step toward a unified theoretical framework was provided by Uttal et al. (2013), who categorized spatial ability along two cognitive dimensions: static versus dynamic and intrinsic versus extrinsic (Figure 1). The static-dynamic axis distinguishes between fixed spatial properties of individual objects and their transformations over time, while the intrinsic-extrinsic axis differentiates between operations within a single object and those concerning multiple objects or environmental contexts. The intersection of these dimensions produces four categories of spatial tasks, offering a structured lens for understanding spatial ability, though primarily within the psychological domain.

Most psychological studies have focused on small-scale spatial abilities—body-centered operations involving limited extents, such as object rotation or internal visualization—without fully addressing how people understand geographic environments. These abilities are commonly assessed through tasks such as the Purdue Spatial Visualization Test (PSVT) (Bodner and Guay 1997) or perspective-taking tasks like the Perspective Taking/Spatial Orientation Test (PTSOT) (Hegarty 2004; Friedman et al. 2020). In more recent years, however, researchers have increasingly recognized the importance of spatial thinking in geographic contexts, including navigation, map interpretation and regional understanding (Kastens and Ishikawa 2006; Wakabayashi and Ishikawa 2011; Ishikawa 2013; Wakabayashi 2015). Kozhevnikov and Hegarty (2001) and Hegarty et al. (2006) demonstrated that spatial abilities related to geographic environments and those assessed in laboratory settings are both important but need to be studied independently. Jansen (2009) further showed

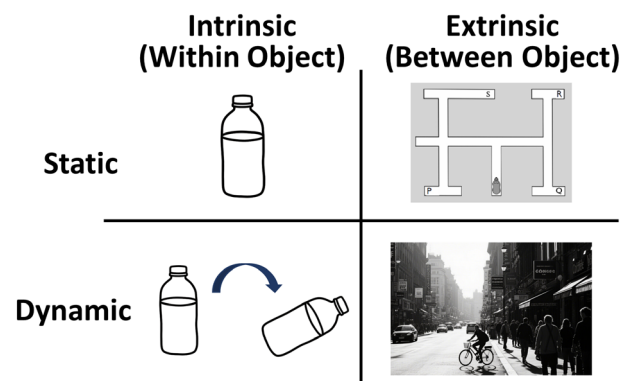


FIGURE 1 | Two dimensions of spatial ability (modified from Uttal et al. 2013).

that, while correlations exist in adults, children's performance on psychological measures of spatial ability (e.g., object manipulation) did not predict their spatial skills in a geographic context (e.g., wayfinding). These findings highlight geospatial competencies involve broader contextual understanding, distinct from those captured in traditional psychological assessments (Bednarz and Lee 2012; Ishikawa 2013; Dong et al. 2023).

To evaluate these varied dimensions, researchers have developed a wide range of tests in both cognitive psychology and geography. Psychological assessments typically measure individual operations in constrained spaces. For instance, the PVRT evaluates mental rotation, while the PTSOT measures the ability to mentally shift perspectives. By contrast, geography-oriented assessments address higher-order geospatial tasks. The Spatial Thinking Ability Test (STAT) (Bednarz and Lee 2012) evaluates spatial orientation, information comparison, and spatial decision-making. The Spatial Skills Test (SST) (Jongwon Lee and Bednarz 2009) tracks learning progress following GIS instruction, while Topographic Map Assessment (TMA) (Newcombe et al. 2015) measures the ability to interpret topographic maps.

In summary, traditional research on spatial ability has been largely rooted in the field of cognitive psychology, producing unified frameworks and standardized tests for evaluating spatial skills. While these contributions provide a strong cognitive foundation, they often overlook the broader demands of spatial thinking in geographic contexts. As spatial reasoning becomes increasingly vital in education, GIS, and artificial intelligence (AI), there is a pressing need to expand psychological frameworks to capture the complexity and contextual richness of geographic spatial problems.

2.2 | Evaluation of Spatial Ability in LLMs

Research on evaluating spatial ability in large language models (LLMs) can be broadly divided into two lines of inquiry: text-based spatial reasoning and vision-augmented spatial evaluation.

The first line examines how well LLMs perform spatial tasks using natural language alone. For example, Mooney et al. (2023) assessed the reasoning of ChatGPT-3.5 and ChatGPT-4 in GIS contexts and found that although the model produced plausible answers, it often failed to incorporate real-world spatial constraints. Hu et al. (2023) investigated spatial description extraction from disaster-related social media messages in GPT-2, GPT-3, ChatGPT and GPT-4, showing that performance improved when geospatial cues were explicitly embedded. Li, Yao, et al. (2024) introduced a benchmark of 13 spatio-temporal reasoning tasks, concluding that while LLMs, such as GPT-4o, ChatGLM2/3, DeepSeek-7B and etc., handle simple spatial categories, they struggle with topological and geometric reasoning. Other work, such as Ji et al. (2025) and Yang, Fu, et al. (2025), emphasized the role of prompt engineering and spatial context augmentation in enhancing performance on spatially grounded tasks. A subset of this research evaluates

code-based problem solving, where LLMs generate geospatial code (e.g., SQL, Python, GIS scripting). Xu, Zhao, et al. (2025) tested GPT-3.5-turbo, GPT4-turbo, GPT-4o, Claude-3-sonnet, Moonshot-v1-8k and GLM-4 on 90 geospatial coding problems and revealed a notable gap between general-purpose models and domain-specific requirements. Similarly, Hou et al. (2025) proposed GeoCode-GPT, which uses structured prompts and geospatial task abstraction modules to better align model outputs with domain needs.

The second line extends evaluation to multimodal settings by incorporating visual or map-based inputs. Tao and Xu (2023) and Xu and Tao (2024) analyzed GPT-4V's performance in interpreting choropleth maps and infographics, revealing that it often relied on superficial visual cues rather than deeper spatial logic. Danish et al. (2025) introduced a benchmark for evaluating visual-language models on tasks like land use detection and spatial alignment, reporting substantial performance variation among models like GPT-4o, Qwen2-vl, and LLaVA-OneVision. Huang et al. (2025) further examined visual cognition of GPT-4o, Gemini-1.5-pro, Gemini-2.0-pro, and Qwen-VL-Max across primitives such as color, shape, and position, emphasizing persistent weaknesses in spatial consistency and object localization.

While computer vision researchers have advanced AI's spatial intelligence in human-scale 3D environments (e.g., object detection in indoor scenes), and companies develop world models for robotic navigation, these efforts primarily address egocentric, metric 3D spaces at meter-to-room scales. For instance, Mangalam et al. (2023) evaluate long-term, indoor egocentric reasoning on 5000 three-minute video clips. Yang, Yang, et al. (2025) test MLLMs, such as GPT-4o, Gemini-1.5 Flash, InternVL2-2B, and etc., on single-room metric tasks, such as object counts, relative distances, and route plans, using 288 scanned indoor videos. Google's DreamerV3 (Hafner et al. 2023) learns to mine diamonds in Minecraft over hundred-meter horizons. Meta's Navigation World Model (Bar et al. 2025) imagines 16-s future frames with a 1 B-parameter diffusion transformer for corridor- to garden-scale navigation. All of these advances remain anchored in egocentric and metric 3D reconstruction and immediate action selection. In contrast, this paper focuses on geographic-scale spaces, ranging from urban neighborhoods to regional extents, emphasizing topological relationships and GIS workflows.

Beyond benchmarking, several studies have proposed a hybrid framework to address these limitations. Yang, Fu, et al. (2025) introduced Hybrid Mind, which transforms qualitative LLM outputs into structured spatial representations and integrates GIS toolchains for downstream analysis. Xu, Yu, et al. (2025) developed RS-Agent, a modular system that combines LLMs with remote sensing toolkits and spatial knowledge bases, demonstrating improved performance in workflows such as change detection and map classification. Other frameworks, including MapReader (Zhang, Zhang, et al. 2025) and MapColorAI (Yang, Wang, et al. 2025), target cartographic contexts by teaching models visual-symbolic associations. These works show that pretraining on map-specific datasets enhances accuracy in layout and color understanding.

In summary, while recent research has made significant strides in evaluating and augmenting spatial abilities in LLMs, most existing studies remain task-driven and lack a comprehensive, cognitively grounded framework, highlighting the need for more principled and psychologically informed evaluation paradigms.

3 | Methods

We designed the GVSABench that extends these dimensions into geographic contexts (Section 3.2). Based on this design, we constructed a benchmark dataset (Section 3.3) and conducted a comprehensive model evaluation (Section 3.4). The overall

framework is illustrated in Figure 2, with each stage described in the following subsections.

3.1 | Construction of GVSABench Test Dimensions

3.1.1 | GVSA Conceptualization

We extend the framework proposed by Uttal et al. (2013) by integrating the geographic contexts to establish a more comprehensive taxonomy of GVSA (Figure 3). Uttal et al.'s framework classifies spatial tasks along two cognitive dimensions: static versus dynamic and intrinsic versus extrinsic (Section 2.1), which has been widely adopted in previous studies (W. Yang et al. 2020;

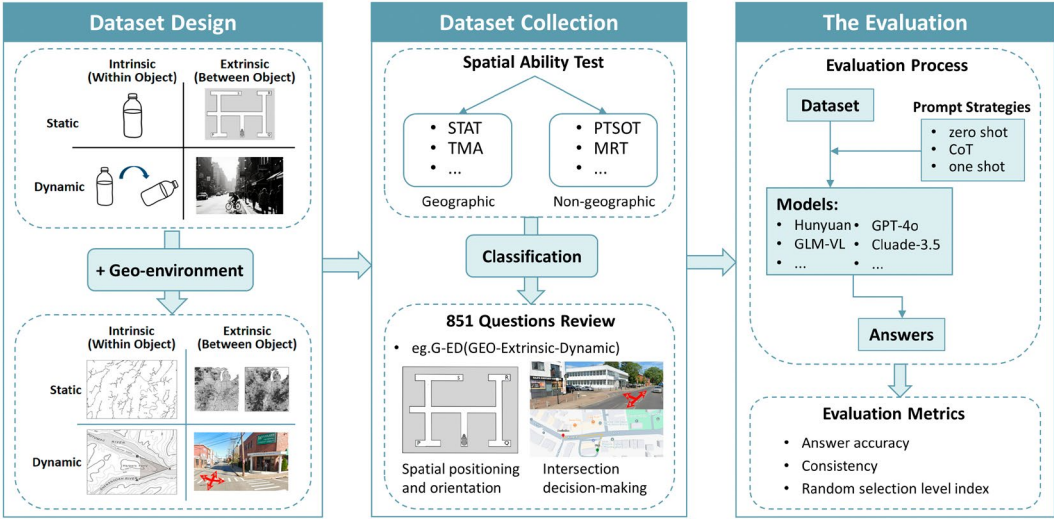


FIGURE 2 | Technical framework of the GVSABench.

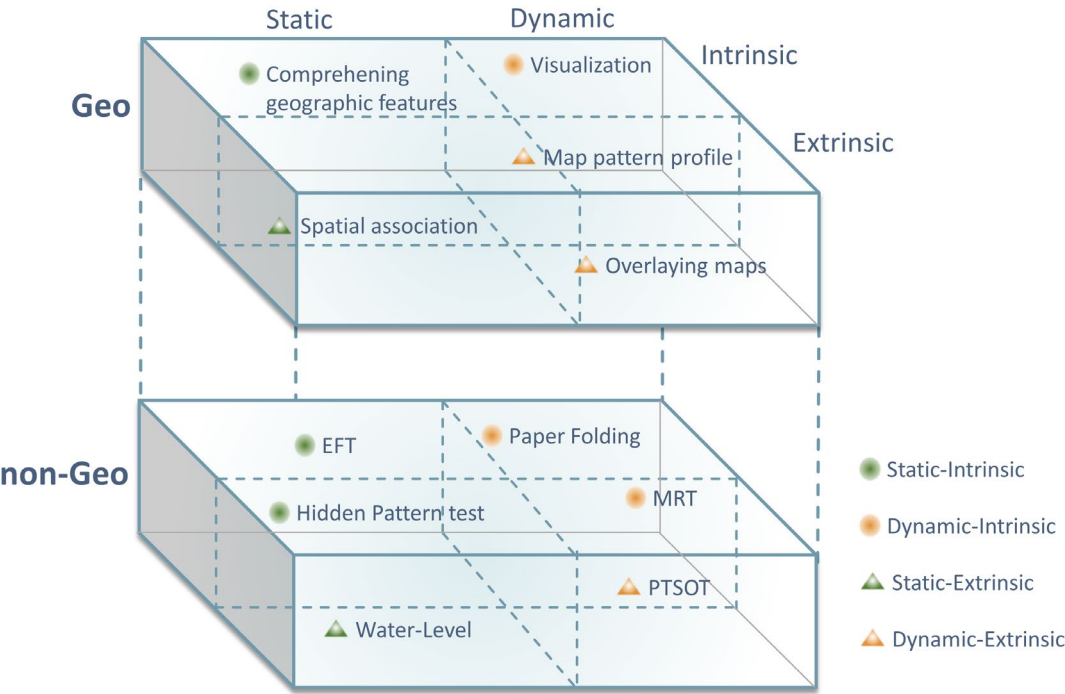


FIGURE 3 | Visuospatial ability test dimensions.

C. Zhu et al. 2023; Schenck and Nathan 2024). We introduce a third dimension: geographic versus non-geographic (Geo vs. non-Geo). This dimension encompasses tasks that involve broader environmental or geographic contexts. The framework is defined as follows:

1. The *static* versus *dynamic* axis describes whether the spatial features are fixed (e.g., object shape) or involve change and movement over time. Static tasks involve invariant spatial information, such as object properties or internal part-whole relationships. Dynamic tasks require reasoning about spatial changes over time, such as movement, transformation, or rotation.
2. The *intrinsic* versus *extrinsic* axis describes the absolute-ness and relativity of objects. Intrinsic tasks focus on spatial information within a single object, while extrinsic tasks involve relations between multiple objects or between objects and their environment.
3. The *geographic* (Geo) versus *non-geographic* (non-Geo) axis describes whether the spatial perception, cognition, and reasoning involve broader environmental or geographic contexts. A key distinction between Geo versus non-Geo is the underlying scale. While non-Geo typically refers to operations in a desktop or smaller than a room space, such as the manipulation and imagination of a square or a cup of water, Geo involves large-scale environments (districts, regions, cities, countries, or global scales), requiring reasoning about layouts, distributions of phenomena, or embodied interactions such as wayfinding.

By adding the geographic dimension, Uttal et al.'s original two-dimensional framework expands into a three-dimensional cube (Figure 3). This cube enables more nuanced classification of spatial tasks, systematically extending existing categories into geospatial environments and facilitating the design of tasks that reflect real-world geographic cognition.

3.1.2 | Task Categories

Combining the three dimensions yields eight distinct categories of GVSA (Table 1). These include the four original non-geographic categories and their geographic extensions. Large-scale geographic problems are more complex, often requiring different cognitive strategies than small-scale tasks. For example, mentally rotating a cube may involve a fixed coordinate system, whereas interpreting a topographic map often requires imagining movement within a geographic coordinate system. Following this principle, we derive four new geographic task categories in parallel with the non-geographic ones. Table 1 summarizes the categories and their associated task types. A detailed description of each category and example questions are included in the Appendix 1.

3.2 | Testing Dataset Collection and Design

3.2.1 | Non-Geo Tests

For non-Geo tests, a wide range of standardized assessment tools already exists. These tools have been extensively used to evaluate specific dimensions of spatial ability. For this category, we primarily drew from established resources and incorporated representative tasks from several well-validated instruments, including the Factor-Referenced Cognitive Test (FRCT) (Ekstrom et al. 1976), the PTSOT (Hegarty 2004; Friedman et al. 2020), the PSVT (Bodner and Guay 1997), the Cross Section Test (CST) (Cohen and Hegarty 2012), and the Mental Rotation Test (MRT) (Peters et al. 1995).

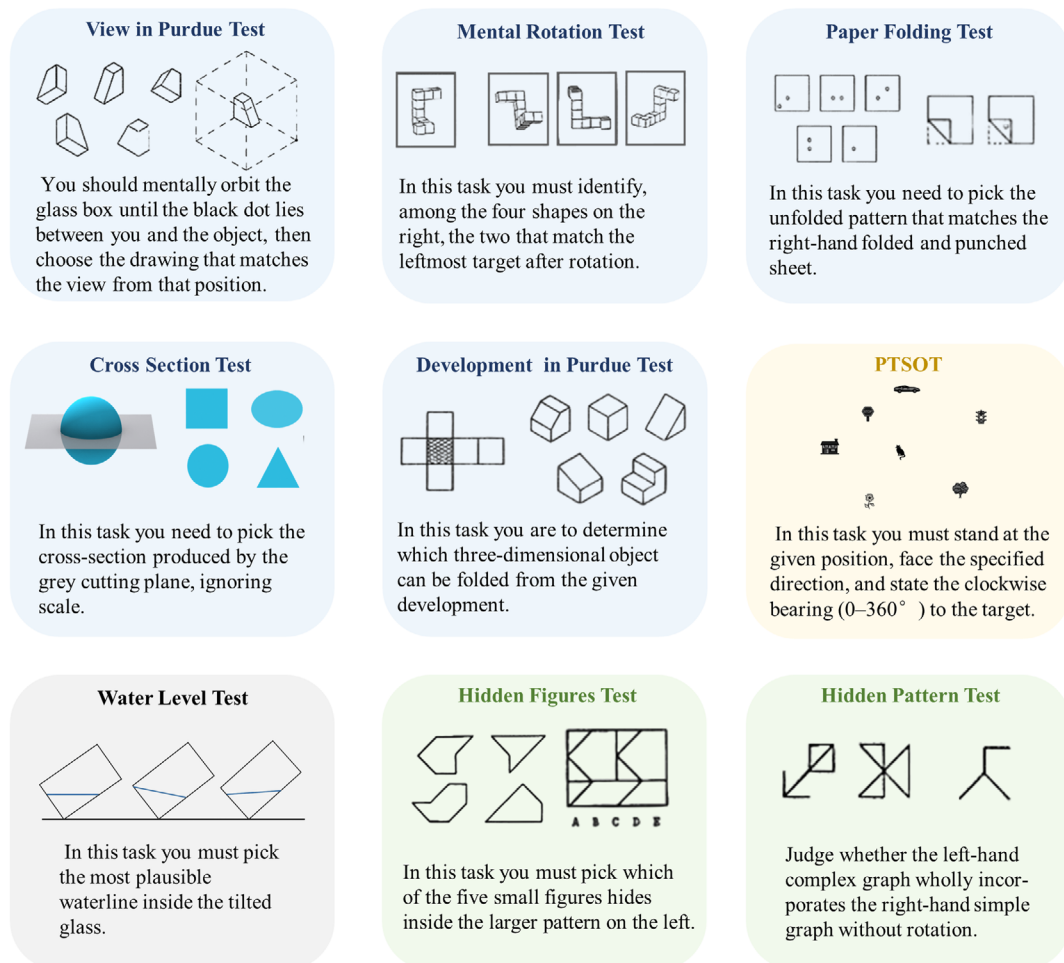
To ensure suitability for LLM-based responses, only selected items were adopted. From the FRCT, we included the Hidden Figures Test, Hidden Patterns Test, Cube Comparison, and Paper Folding Test, as these are most relevant to visuospatial operations. For the PSVT, only a subset of items was retained due to redundancy across tasks. In contrast, the remaining test batteries were

TABLE 1 | Task categories of geo-visuospatial ability.

Category		Description
Non-geographic (NG)	Intrinsic-static (IS)	Perceiving objects, paths, or spatial configurations amid distracting background information
	Intrinsic-dynamic (ID)	Constructing complex configurations from parts, visualizing and mentally transforming objects, or rotating 2D/3D objects
	Extrinsic-static (ES)	Grasping abstract spatial concepts, such as horizontal invariance or verticality
	Extrinsic-dynamic (ED)	Visualizing an entire environment from different perspectives
Geographic (G)	Intrinsic-static (IS)	Recognizing and interpreting spatial information in a geographic context, such as identifying terrain features
	Intrinsic-dynamic (ID)	Processing dynamic geospatial information, including constructing or deconstructing geospatial elements and translating between 2D topographic maps and 3D terrain
	Extrinsic-static (ES)	Applying geographic principles to recognize relative positions and layouts of features and to infer spatial relationships
	Extrinsic-dynamic (ED)	Navigating and orienting within a geographic environment, adapting to changes in spatial perspective and reference systems

TABLE 2 | GVSA test distribution.

Category		Abbreviation	Tests	Question count
Non-geographic	Intrinsic-static	NG-IS	Hidden Figures Test; Hidden Patterns Test	132
	Intrinsic-dynamic	NG-ID	Cube comparison; Mental Rotation Test; Purdue Test; Paper Folding Test; Cross Section Test	387
	Extrinsic-static	NG-ES	Water Level	20
	Extrinsic-dynamic	NG-ED	Perspective Taking Spatial Orientation Test	32
Geographic	Intrinsic-static	G-IS	Comprehending geographical features	70
	Intrinsic-dynamic	G-ID	Spatial construction; Spatial deconstruction	70
	Extrinsic-static	G-ES	Spatial localization; Spatial relation	70
	Extrinsic-dynamic	G-ED	Intersection decision-making; Map overlay; Spatial orientation	70
Total				851

**FIGURE 4** | Dataset overview of non-Geo tasks. Brief task descriptions are shown below each example image (not the prompts for model evaluation).

included in their entirety. A total of 571 questions were included in the non-Geo category (Table 2). Examples are shown in Figure 4.

3.2.2 | Geo Tests

For Geo tests, we collected items from the STAT (Bednarz and Lee 2012), TMA (Newcombe et al. 2015), self-location and spatial orientation (Liao et al. 2022), Geospatial Thinking Test (Jiang 2018), which assesses geospatial thinking skills such as mental manipulation of geospatial information and understanding of geospatial relationships. This resulted in 102 items. The number of items in Geo tests is substantially smaller than that in non-Geo tests. To mitigate dataset imbalance, we manually designed 120 new items, each aligned with the distinctive features of its category. For instance, we utilized Google Street View to design tasks such as intersection decision-making and spatial localization. Furthermore, we leveraged the MLLM Seedream 4.0 (Seedream et al. 2025) to generate artificial street images and thematic maps, which assisted in compiling 80 questions items (details in Section 5.7). A total of 280 questions were included in this type (Table 2). The compiled tests for Geo tasks are shown in Figure 5.

3.3 | Dataset Processing

After collection, all tasks were transformed into formats suitable for evaluation by LLMs. Most original tests contained four components: (1) task instructions provided to participants, (2) example items for comprehension, (3) test images as visual stimuli, and (4) corresponding correct answers. To adapt them for LLM-based evaluation, we integrated these components into the LLM evaluation process. Specifically, *the task instructions* were

reformulated into natural language prompts to clearly convey the task objectives to the model. *The example items* were optionally included as part of one-shot prompting to help the model understand the response format and reasoning process. *The test images* were used as the visual inputs for multimodal models. Finally, *the correct answers* were used as ground truth labels for automated scoring.

Three main answer formats were included: (1) multiple-choice questions, accounting for over half of the items, requiring one or two correct selections from several options; (2) true-or-false questions, used only in non-geographic tasks, requiring binary judgments; and (3) objective questions, used sparingly in geographic tasks, requiring definite numerical or textual responses. Together, these formats approximate the variety of response types used in human assessments.

3.4 | The Evaluation

All evaluations were conducted by programmatically invoking the Application Programming Interfaces (APIs) of each model under three prompting strategies and within two language environments (Chinese and English). The tested models, including three English-based models (GPT-4o, Gemini-2.0, Claude-3.5-Sonnet) (OpenAI et al. 2024; Anthropic 2024) and four Chinese-based models (Qwen-VL-Plus, GLM-4V-Plus, Hunyuan-vision, and Deepseek-VL2; X. Sun et al. 2024; Wu et al. 2024), are listed in Table 3. These models cover a representative set of state-of-the-art MLLMs and were latest available during our experimentation period in January 2025, offering both text and image input capabilities via public APIs. This selection aligns with prior related studies focusing on these or similar models (Hochmair et al. 2024; Hou et al. 2024;

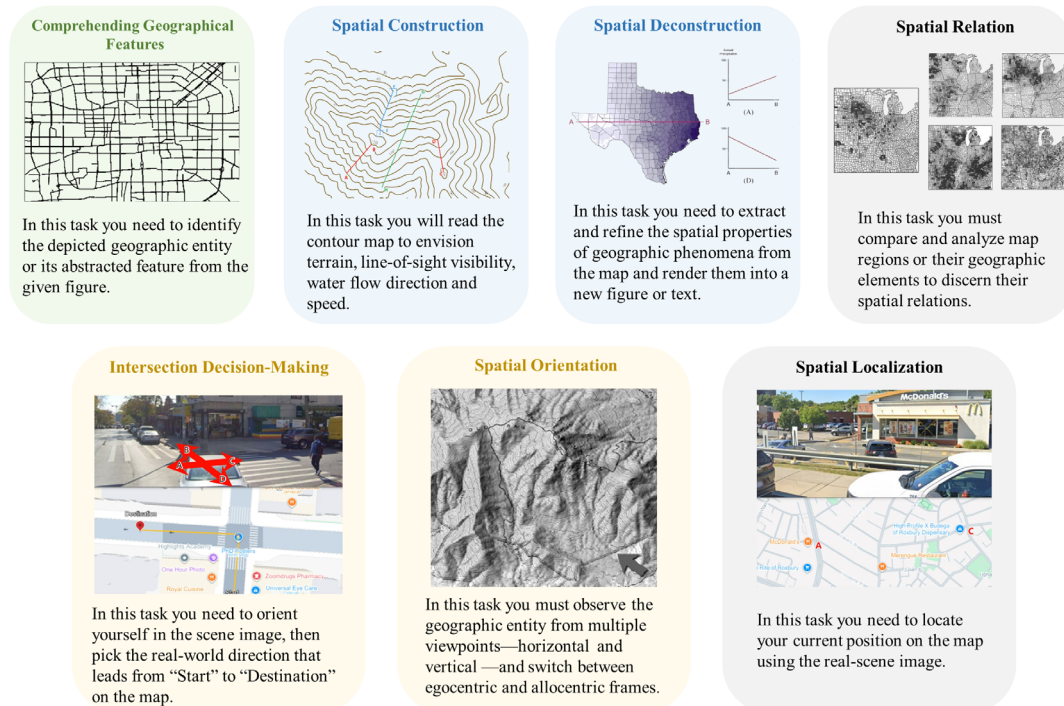


FIGURE 5 | Dataset overview of Geo tasks. Brief task descriptions are shown below each example image (not the prompts used for model evaluation).

Ji et al. 2025; Yang, Fu, et al. 2025), ensuring relevance and comparability. Additionally, these models supported multimodal API access, while other models such as ERNIE (Y. Sun et al. 2021) did not provide image-input APIs during our experiment period, limiting their applicability.

Three prompt strategies were employed: zero-shot (no additional context), one-shot (minimal demonstration), and zero-shot Chain-of-Thought (CoT) reasoning. Zero-shot CoT is divided into simple CoT (SCoT) and complex CoT (CCoT). Examples are shown in Table 4. During evaluation, both textual and visual inputs were submitted to the models through API calls, and the outputs were automatically collected for subsequent analysis. Illustrative input–output examples using the Hunyuan-Vision model were provided in Figures 6 and 7.

To minimize potential biases from prior interactions and ensure that each model's output reflected an independent reasoning process, we adopted a single-turn dialogue setting in which each input prompt was submitted separately without any historical conversation context. In addition, to examine the model consistency, we performed two independent rounds of testing for each sample while keeping the temperature parameter fixed (i.e., no randomness introduced), thereby controlling for stochastic variation across runs.

We used accuracy and consistency to measure model performance. Accuracy is defined as the proportion of correct responses out of the total number of questions. The accuracy was normalized to 0 to 100. Consistency is applied to evaluate the model's stability when repeatedly tested with the same question.

TABLE 3 | Seven multimodal large language models used in the test.

Model	Model version	Open-source	Company	Link
Qwen	qwen-vl-plus-0102	Yes	Alibaba	https://chat.qwen.ai/
Hunyuan	Hunyuan-vision	No	Tencent	https://hunyuan.tencent.com/
GLM	GLM-4V-Plus-0111	No	Zhipu	https://www.zhipuai.cn
Gemini	Gemini-2.0-flash-001	No	Google	https://cloud.google.com/vertex-ai/generative-ai/docs
Claude	Claude-3.5-sonnet	No	Anthropic	https://claude.ai/
GPT	gpt-4o-2024-11-20	No	OpenAI	https://chatgpt.com/
Deepseek	deepseek-vl2	Yes	Deepseek	https://www.deepseek.com/

TABLE 4 | Prompt strategies used in the test.

Prompt strategy	Example
zero-shot	“What is the shortest path from A to B?”
one-shot	“I will provide an example: Question: ‘How do I get from A to B?’ Answer: ‘If you start from A, move downward to 6, then move right through 7, 8, 9, and 10, and finally move upward to reach B, that is one possible path’ Now, I want a shorter path. Please provide an answer based on the example”
zero-shot with simple CoT	“What is the shortest path from A to B? Let's think step by step”
zero-shot with complex CoT	“Task: Find the Shortest Path Actions: 1. Analyze the positions of the start and end points: Determine the exact locations of the starting point A and the endpoint B within the grid, and analyze their relative positions 2. Establish movement rules: Clarify the legal ways to move from one node to another, which are limited to moving up, down, left, or right to an adjacent node 3. Plan the path: Based on the positions of the start and end points and in accordance with the movement rules, plan a path from the starting point to the endpoint. Prioritize direct movements and avoid unnecessary detours 4. Evaluate the path length: Calculate the length of the planned path to ensure that it is the shortest possible path Objective: To find the shortest path from the starting point A (position 1) to the endpoint B (position 5), ensuring that the path is the shortest and complies with the movement rules Now, please provide the answer based on the hints”

```
# hunyuan-vision
completion = client.chat.completions.create(
    model=" hunyuan-vision",
    messages=[{"role": "user", "content": [
        {"type": "image_url",
         "image_url": YOUR_IMAGE_URL_HERE},
        {"type": "text",
         "text": " Select the location closest to the real scene image on the map."}],
    }]]
)
```

FIGURE 6 | An input–output example using the Hunyuan-Vision model.

4 | Results

4.1 | Performance of Zero-Shot

4.1.1 | Accuracy Across Languages, Models, and GVSA Categories

Table 5 presents the overall accuracy of the first round of zero-shot tests conducted in Chinese and English. It is seen that accuracy ranged from 36.62% to 54.84% in Chinese and from 37.86% to 52.36% in English. Gemini achieved the highest accuracy in both languages, with 54.84% in Chinese and 52.36% in English, followed by GPT, Claude, GLM, and Deepseek. Hunyuan ranked seventh, while Qwen performed slightly worse than Hunyuan in both Chinese and English. With the exception of GPT-4o, which achieved higher accuracy in English by 2.95 percentage points, the other models performed similarly across both languages. Hence, the differences between languages were not substantial.

Table 6 and Figure 8 display accuracy across the eight GVSA categories in both languages. Accuracy ranged from 10.0% to 85.0% in Chinese and from 15.0% to 75.0% in English. Remarkably, Gemini achieved an accuracy of 85% in NG-ES tasks, substantially outperforming the other models. Moreover, it secured the top accuracy in seven Chinese sub-tasks and in three of the English sub-tasks. GPT reached 71.4% accuracy in G-ES tasks and 77.1% accuracy in G-IS tasks in English. Notably, in NG-ID, NG-IS, and G-ED tasks, accuracy remained close to or below the random guess, demonstrating limited effectiveness in these visuospatial tasks.

To assess performance discrepancies across languages, the accuracy difference between Chinese and English was analyzed, as shown in Figure 9. A clear separation emerges between non-Geo and Geo categories. Specifically, non-Geo categories (white background) exhibited minimal accuracy differences between languages, with accuracy differences ranging from −25% to 20% (mean = 0.81%). This indicates that model performance is largely stable across languages in small-scale non-Geo contexts. Claude and Deepseek are exceptions, as Claude scored 20% higher in Chinese than in English on NG-ES, while Deepseek's English accuracy on the same task exceeded its Chinese performance by approximately 25%. In contrast, the Geo categories (light green background) showed more pronounced language-related disparities, with differences ranging from −20.0% to 8.3% (mean = −2.6%).

4.1.2 | Consistency

Figure 10 summarizes the proportion and number of inconsistent responses across all tasks. Claude, GPT, Gemini, and Deepseek exhibited both a higher quantity and proportion of inconsistent answers compared to other models. Deepseek, in particular, exceeded 50% inconsistent responses. In contrast, Qwen demonstrated the highest consistency, with only 10% inconsistency. Additionally, Claude stood out for having a higher proportion of inconsistent answers in the G-ID tasks than other models and categories.

4.2 | Prompt Tuning

4.2.1 | Zero-CoT

Table 7 and Figure 11 illustrate the accuracy and accuracy improvement of simple CoT compared to zero-shot. It is seen that in both languages, incorporating a simple “let's think step by step” had minimal effect on accuracy, apart from Qwen under Chinese, where scores in NG-ES rose by 30%, and in some cases even reduced performance. For instance, GPT showed a 30% decline on NG-ES tasks in Chinese, while Gemini dropped 22.9% on G-IS tasks in Chinese.

A more complex zero-CoT prompt using “task, action, and goal” yielded some improvements (Table 8 and Figure 12), particularly in Chinese. For example, Qwen showed a 55% increase in accuracy for the NG-ES tasks. However, improvements were smaller in English. Interestingly, for both languages, under the zero-CoT prompt, models exhibited larger accuracy changes on geographic and NG-ES tasks than on the other non-geographic tasks, indicating heightened sensitivity.

4.2.2 | One-Shot

The one-shot approach, which integrates example input alongside model queries, generally produced minimal improvements (Figure 13). The main exception was Qwen, which improved by 35% on NG-ES tasks in Chinese. However, this was still weaker than the 60% improvement achieved with zero-CoT (see Figure 12). In both languages, the one-shot strategy negatively affected models such as Claude, GPT, and Gemini, particularly

Instruction

You are to: 1. study how the object in the top line of the question is rotated; 2. picture in your mind what the object shown in the middle line of the question looks like when rotated in exactly the same manner; 3. select from among the five drawings (A, B, C, D, or E) given in the bottom line of the question the one that looks like the object rotated in the correct position. No need to explain. The format of the answer should be: The answer is XXX.

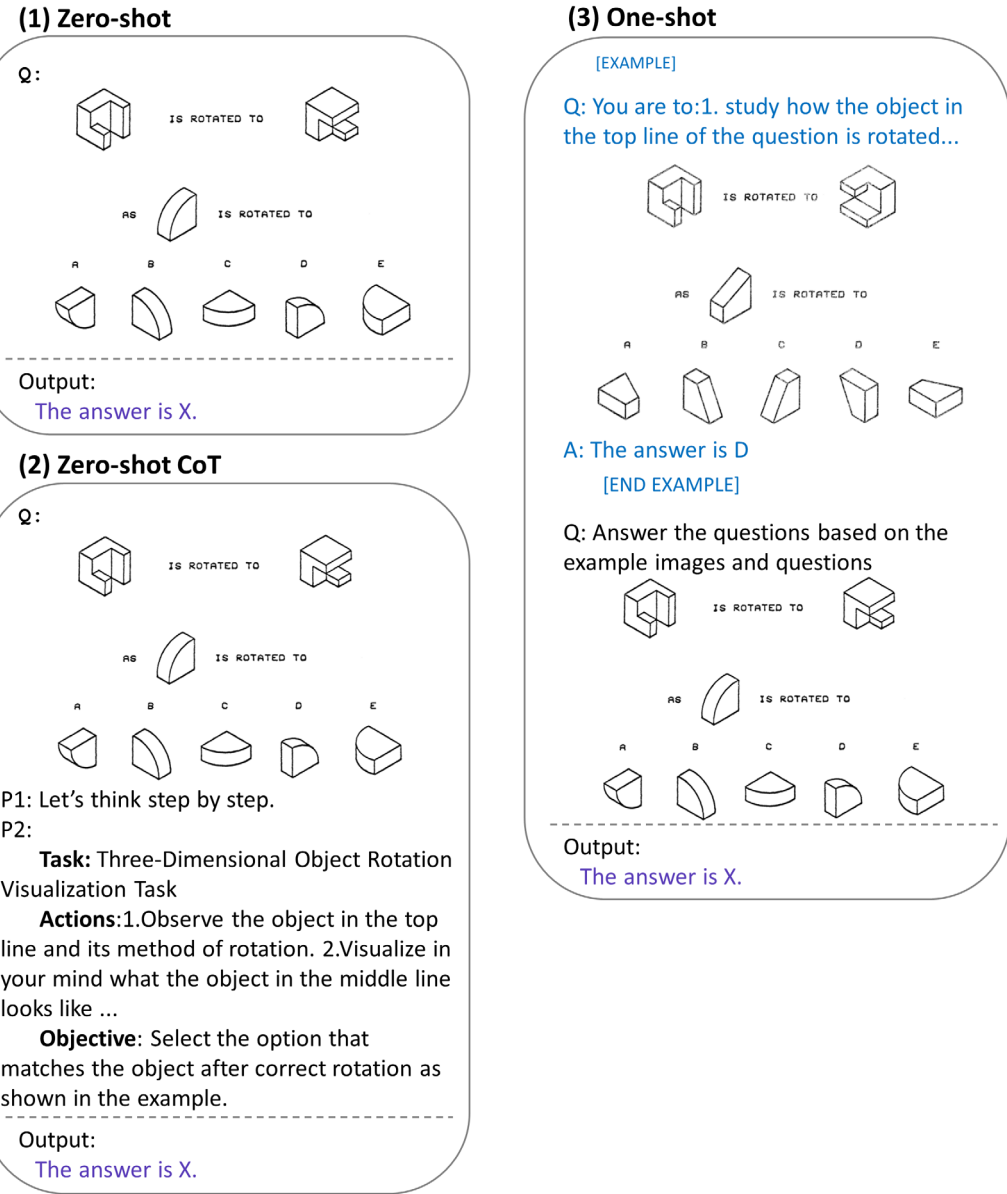


FIGURE 7 | Prompt strategies demonstration.

on geographic tasks. For example, GPT’s accuracy declined by about 50% compared with zero-shot in G-IS and G-ES tasks.

5 | Discussion

5.1 | Comparison to Previous Studies on Model Performance

The results indicate that current models performed suboptimally on GVSA tasks, with zero-shot accuracy typically ranging

from 20% to 80% (random guessing: 13.7%–50.0%) and consistency between 10% and 50%. Gemini-2.0 achieved the highest score of 54.84 in Chinese. These findings are broadly consistent with prior studies reporting accuracy ranges of 1.6%–62.1% for Gemini-2.0-Pro (Huang et al. 2025), 20%–80% for GPT-4v (Zhang, Wang, et al. 2024), and 24.5%–66.7% for Gemini-2.0 Flash (Yang, Fu, et al. 2025). Collectively, these results highlight the substantial limitations of MLLMs in visual cognitive tasks.

Although GPT has demonstrated strong performance in certain spatial tasks, such as spatial literacy, function interpretation,

and code explanation (Borji and Mohammadian 2023; López Espejel et al. 2023; Xu, Zhao, et al. 2025), GPT-4o did not outperform other models in the present study. This may be due to past research focusing primarily on natural language tasks and overlooking multimodal capabilities (Hochmair et al. 2024; Xu, Zhao, et al. 2025; Yang, Fu, et al. 2025). Furthermore, existing studies on multimodal abilities have often been focused

on specific spatial tasks (e.g., map reading, route planning, topographic-map interpretation) or on narrowly defined test scales (Li, Haider, and Callison-Burch 2024; Tao and Xu 2023; Xu and Tao 2024), resulting in an incomplete evaluation of models' broader spatial capabilities in real multimodal interactions that require visual, linguistic, and spatial reasoning.

In summary, our findings are largely consistent with existing studies, collectively reinforcing the conclusion that current multimodal large language models still exhibit limited capacity in spatial understanding, particularly in tasks that require the integration of visual, linguistic, and geographic reasoning. Our results underscore the need for further research to enhance the spatial cognitive abilities of MLLMs, especially in handling complex, real-world geo-visuospatial tasks across varying spatial scales.

5.2 | Analysis of Model Failures in GVSA Tasks

To gain deeper insights into the model's successes and failures, we used Gemini-2.0, the best-performing MLLM on the

TABLE 5 | Overall accuracy of zero-shot in two languages.

Model	Chinese (%)	English (%)
claude-3.5-sonnet	44.50	45.00
gpt-4o-2024-11-20	45.96	48.91
gemini-2.0-flash	54.84	52.36
glm-4v-plus-0111	43.93	44.88
hunyuan-vision	36.66	38.59
qwen-vl-plus-0102	36.62	37.86
deepseek-vl2	41.98	42.88

Note: Bold values represent the best.

TABLE 6 | Accuracy of zero-shot by GVSA category, model and language.

Category	Chinese (%)							English (%)							Random guess (%)
	Cl	Ge	G4	Gl	Hv	Qw	Ds	Cl	Ge	G4	Gl	Hv	Qw	Ds	
NG-ID	39.8	37.7	38.8	43.2	33.3	36.7	38.0	39.8	39.3	36.4	38.0	34.4	38.2	39.0	35.8
NG-IS	43.9	49.2	44.7	42.4	44.7	43.2	43.9	40.9	42.4	46.2	37.9	45.5	43.9	34.8	42.7
NG-ED	48.7	45.8	51.6	51.2	64.6	50.5	62.5	56.4	46.7	46.3	52.6	58.3	46.1	58.5	50.0
NG-ES	55.0	85.0	45.0	20.0	10.0	15.0	40.0	35.0	75.0	45.0	25.0	15.0	20.0	65.0	33.3
G-ID	43.1	50.0	36.1	45.8	26.4	36.1	38.9	37.5	41.7	44.4	52.8	20.8	34.7	34.7	27.1
G-IS	51.4	72.9	70.0	67.1	61.4	65.7	52.9	70.0	78.6	77.1	70.0	70.0	65.7	60.0	25.4
G-ED	25.7	25.7	24.3	27.1	11.4	15.7	18.6	25.7	27.1	17.1	24.3	24.3	18.6	15.7	23.7
G-ES	41.4	64.3	51.4	47.1	37.1	24.3	40.0	48.6	61.4	71.4	50.0	37.1	30.0	34.3	23.8

Note: Bold values represent the best in each category.

Abbreviations: Cl, claude-3.5-sonnet; Ds, deepseek-vl2; G4, gpt-4o-2024-11-20; Ge, gemini-2.0-flash; Gl, glm-4v-plus-0111; Hv, hunyuan-vision; Qw, qwen-vl-plus-0102.

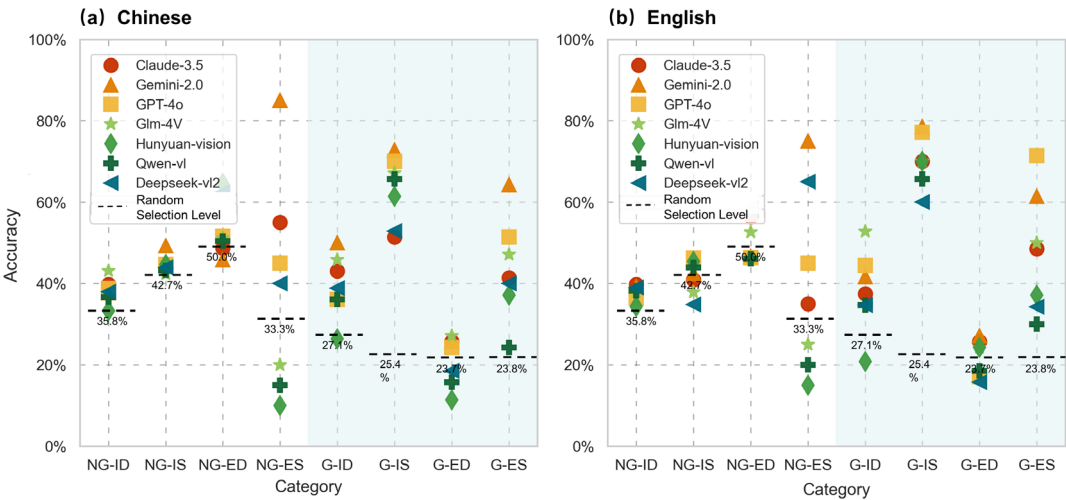


FIGURE 8 | Accuracy by GVSA category in Chinese (left) and English (right).

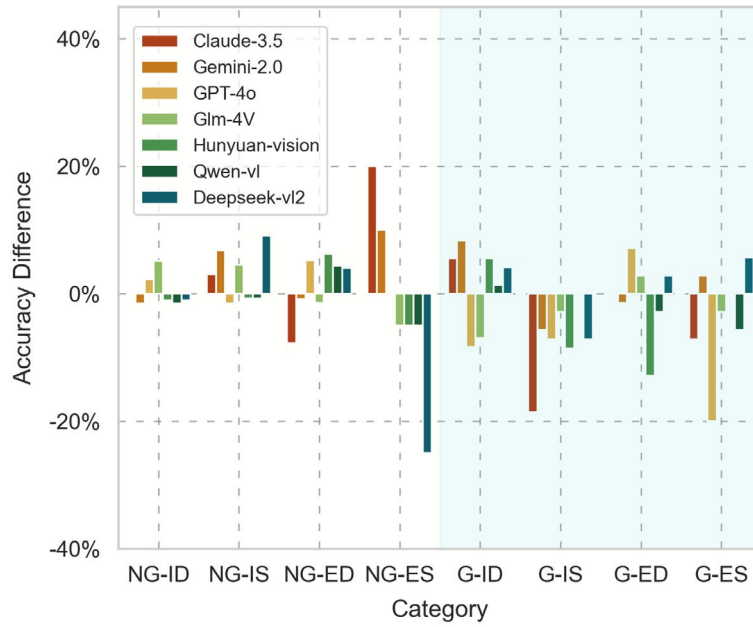


FIGURE 9 | Accuracy differences between Chinese and English (Chinese–English).

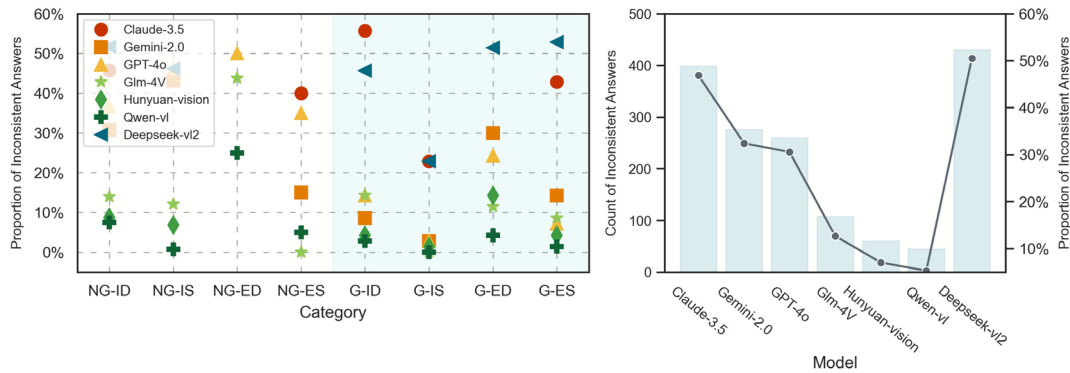


FIGURE 10 | Models' inconsistent response rate (left) and count statistics (right).

TABLE 7 | Accuracy of simple CoT by GVSA category, model and language.

Category	Chinese (%)							English (%)							Random guess (%)
	Cl	Ge	G4	Gl	Hv	Qw	Ds	Cl	Ge	G4	Gl	Hv	Qw	Ds	
NG-ID	37.0	37.7	32.0	44.7	36.4	36.4	36.7	37.2	38.0	37.5	42.6	33.6	38.2	42.1	35.8
NG-IS	41.7	41.7	44.7	38.6	41.7	42.4	42.4	43.2	47.7	46.2	38.6	47.0	38.6	29.5	42.7
NG-ED	51.5	48.4	57.8	56.7	68.1	58.5	61.3	64.2	46.1	57.8	56.0	66.7	65.9	61.8	50.0
NG-ES	60.0	85.0	15.0	15.0	10.0	45.0	30.0	20.0	85.0	30.0	15.0	15.0	15.0	55.0	33.3
G-ID	40.3	37.5	30.6	47.2	31.9	29.2	31.9	33.3	33.3	43.1	59.7	23.6	31.9	26.4	27.1
G-IS	48.6	50.0	68.6	64.3	62.9	64.3	54.3	61.4	67.1	77.1	70.0	71.4	67.1	64.3	25.4
G-ED	20.0	21.4	30.0	24.3	18.6	18.6	20.0	21.4	27.1	28.6	14.3	20.0	25.7	18.6	23.7
G-ES	48.6	52.9	54.3	52.9	41.4	37.1	30.0	50.0	50.0	71.4	55.7	34.3	30.0	34.3	23.8

Note: Bold values represent the best in each category.

Abbreviations: Cl, claude-3.5-sonnet; Ds, deepseek-vl2; G4, gpt-4o-2024-11-20; Ge, gemini-2.0-flash; Gl, glm-4v-plus-0111; Hv, hunyuan-vision; Qw, qwen-vl-plus-0102.

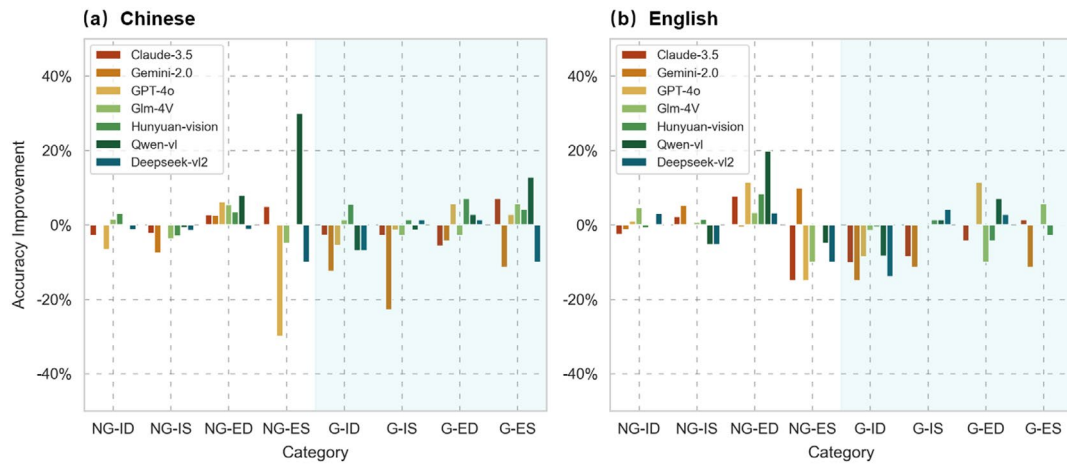


FIGURE 11 | Accuracy improvement of simple CoT compared to zero-shot under Chinese (left) and English (right).

TABLE 8 | Accuracy of complex CoT by GVSA category, model and language.

Category	Chinese (%)							English (%)							Random guess (%)
	Cl	Ge	G4	Gl	Hv	Qw	Ds	Cl	Ge	G4	Gl	Hv	Qw	Ds	
NG-ID	39.0	34.9	38.8	43.7	35.9	38.0	38.2	38.8	40.6	38.5	42.9	34.4	39.0	37.7	35.8
NG-IS	42.4	40.9	43.2	40.2	47.0	38.6	47.0	48.5	45.5	47.7	39.4	49.2	38.6	42.4	42.7
NG-ED	45.3	49.2	45.8	59.0	64.3	51.8	51.9	53.9	50.7	50.4	50.0	61.9	47.3	62.4	50.0
NG-ES	35.0	100	40.0	25.0	10.0	70.0	35.0	55.0	95.0	25.0	40.0	10.0	30.0	50.0	33.3
G-ID	36.1	51.4	38.9	43.1	27.8	37.5	40.3	43.1	44.4	44.4	54.2	20.8	40.3	31.9	27.1
G-IS	60.0	75.7	74.3	67.1	62.9	70.0	54.3	68.6	80.0	78.6	71.4	71.4	61.4	64.3	25.4
G-ED	31.4	40.0	25.7	22.9	15.7	25.7	18.6	12.9	14.3	21.4	35.7	24.3	21.4	12.9	23.7
G-ES	62.9	61.4	54.3	51.4	34.3	25.7	37.1	48.6	62.9	77.1	51.4	40.0	34.3	34.3	23.8

Note: Bold values represent the best in each category.

Abbreviations: Cl, claude-3.5-sonnet; Ds, deepseek-vl2; G4, gpt-4o-2024-11-20; Ge, gemini-2.0-flash; Gl, glm-4v-plus-0111; Hv, hunyuan-vision; Qw, qwen-vl-plus-0102.

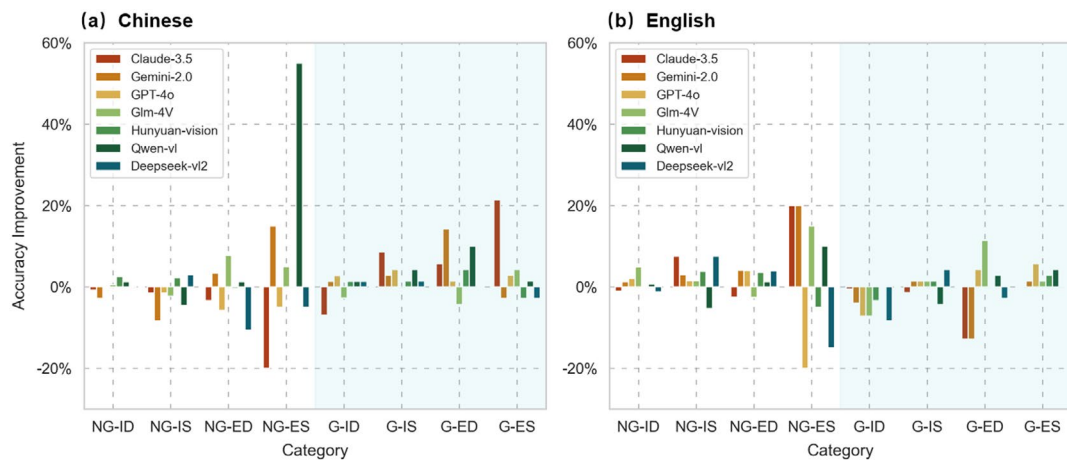


FIGURE 12 | Accuracy improvement of complex CoT compared to zero-shot under Chinese (left) and English (right).

dataset, to re-attempt the questions from mental cutting, intersection decision-making, spatial relation, and MRT subtasks. The model was prompted to articulate its internal reasoning, and the responses were manually reviewed to identify patterns in performance. Figure 14 presents two representative cases.

In the spatial relation task (left panel of Figure 14), the model produced the correct response along with a coherent reasoning chain. This suggests that the model grasped the task logic and engaged in plausible, human-like reasoning. Its explanation, "... especially the clusters of darker and lighter areas in

the same relative locations,” demonstrates visual perception capability, as the model successfully detected grayscale variations, interpreted geographic information, and selected the correct answer.

In contrast, in the intersection-decision task (right part of Figure 14), the model failed. Its reasoning began with “The map shows the ‘Start’ point at the corner of Havemeyer St and a street with ‘Echo Drugs’ on the corner,” when in fact the “Start” point is located far from that corner. This reveals a limitation in visual perception. Further statements, such as “we are at the ‘Start’ point” and “... because the street sign to the right matches the direction on the map!”, indicate an attempt at landmark-based reasoning, but ultimately an incorrect answer (D) was chosen. This failure primarily arose from misalignment between egocentric and allocentric reference frames, with the model mistaking a right turn from a first-person perspective for the correct direction in a map-based layout.

Reviewing all cases in this manner allowed us to identify three major factors underlying MLLM failures:

1. Restricted visual perception. The model often failed to register or correctly identify critical visual details. Consequently, even when the reasoning was logically sound, the final answer was incorrect. For example, most errors in the Cross Section Test stemmed from misjudging the tilt of the interface, while errors in MRT tasks typically arose from failing to recognize abstract block configurations.
2. Difficulty with viewpoint transformation and spatial cognition. Performance was especially weak in intersection-decision and MRT tasks, where reconciling egocentric and allocentric frames proved challenging.
3. Misinterpretation of linguistic information. The model occasionally misunderstood the question wording or misread

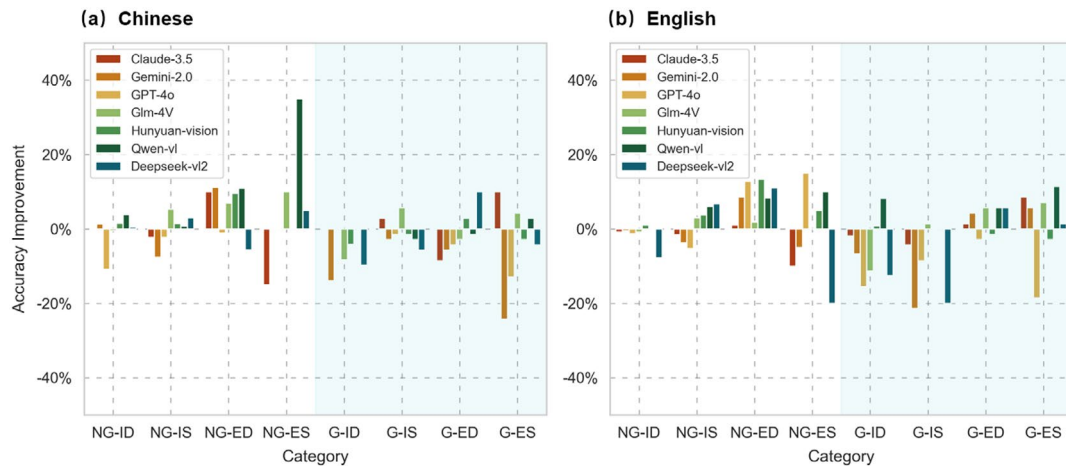


FIGURE 13 | Accuracy improvement of one-shot under Chinese (left) and English (right).

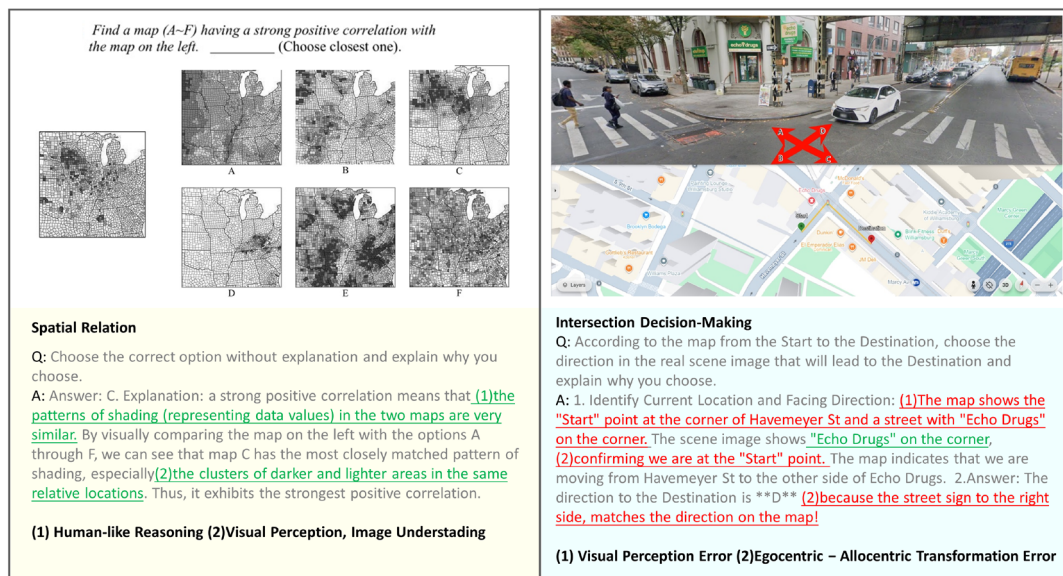


FIGURE 14 | Examples of the model reasoning chains of a success (left) and a failure (right).

text embedded in images, leading to errors. For instance, in the mental cutting task, it frequently failed to follow the instruction “Imagine that you are facing the cutting plane head-on, as if you were looking in a mirror.” As a result, the model often did not visualize the cross-sectional shape from a perpendicular viewpoint.

5.3 | Effectiveness of Prompt Strategies

This study employed two prompt strategies, zero-shot CoT and one-shot, with CoT further divided into simple and complex variants. Average improvement statistics are presented in Table 9. Overall, the prompt strategies produced only slight improvements for some models and negative impacts for others. Among them, the complex CoT yielded the most consistent gains. With the exception of minor decreases in GPT-4o (−1.01%) and Deepseek-VL2 (−1.54%), complex CoT provided modest enhancements across models. The results suggest that the prompting strategies evaluated in this study have limited effectiveness in improving spatial reasoning ability.

These findings are consistent with prior work. For instance, in VISFACTOR tests (Huang et al. 2025), state-of-the-art MLLMs such as GPT-4o, Gemini-Pro, and Qwen-VL performed close to random guessing. Even advanced prompting techniques, including CoT and multi-agent debates (MAD), produced little improvement. Similarly, other studies have reported that strategies such as one-shot, zero-CoT, self-consistency, and Tree of Thoughts (ToT) either failed to enhance or even reduced MLLM performance (Y. Zhang, Bai, et al. 2024; Yang, Yang, et al. 2025).

In general, CoT prompts outperformed one-shot prompts. While the former modestly improved accuracy, the latter sometimes caused performance declines. This disparity likely reflects the models’ greater proficiency with textual information than with visual inputs. Previous research has shown that models relying solely on text often outperform those incorporating visual cues (Y. Zhang, Bai, et al. 2024). For some visual questions, answers can be derived from text descriptions or training data memory without relying on images (Chen, Li, et al. 2024; Zhang, Jiang, et al. 2024). For example, complex CoT increased Gemini-2.0 accuracy by 2.86% in Chinese and 1.81% in English relative

to zero-shot, whereas one-shot reduced accuracy by 5.23% in Chinese. Moreover, under one-shot prompt, more than half of the models exhibited performance degradation (Table 9). Across tasks such as water level, best-location, spatial relation, comprehending geographical features, and spatial construction/deconstruction, accuracy in English declined by up to 20% (Figure 13). This may be attributed to the inclusion of multiple example images, which overwhelmed the model’s attention mechanisms (Zhang, Cheng, et al. 2025). In intersection decision-making tasks, GPT-4o’s reliance on examples was particularly evident: under one-shot prompting, it frequently reproduced answers identical to the examples, leading to accuracy losses. This suggests an overreliance on exemplar matching rather than genuine semantic and contextual understanding (Chen, Li, et al. 2024).

However, even under CoT prompting, significant issues remained. For example, Claude-3.5’s accuracy on NG-ES questions dropped by 20% (Figure 12b), likely due to the limitations of purely text-based reasoning. Text alone cannot capture the nuances of visual information, leading to information loss (Hu et al. 2024). For complex tasks involving “geo-environment” concepts, textual cues were insufficient, and visual grounding was essential for accurate reasoning (Y. Zhang, Bai, et al. 2024).

A comparison of CoT strategies revealed that complex, reasoning-intensive prompts outperformed simpler ones on several tasks. For example, Qwen-VL showed superior performance on water-level tasks in Chinese, while Gemini achieved higher accuracy on water-level, spatial deconstruction/construction, intersection decision-making, and navigation tasks. These gains are attributable to the structure of complex CoT, which explicitly clarifies tasks, actions, and goals, thereby providing targeted guidance and reasoning scaffolding. In contrast, simple CoT prompts were overly generic and failed to direct the model effectively, resulting in weaker visual-spatial processing.

MLLMs responded inconsistently to prompt engineering. As shown in Table 9, Hunyuan-Vision and Qwen-VL gained under all three strategies, whereas GPT-4o and Deepseek-VL2 showed negative gains. Claude-3.5, Gemini-2.0 and GLM-4V achieved the best enhancement with Complex COT, while only GLM-4V, Hunyuan-Vision and Qwen-VL profited from one shot. This reflects differences in model training architectures, resulting in

TABLE 9 | The average improvement in accuracy (%) after using prompt strategies.

Prompt strategy	Language	Model						
		Cl	Ge	G4	Gl	Hv	Qw	Ds
simple CoT	Chinese	−0.19	<u>−7.01</u>	−3.61	−0.04	2.75	5.54	−3.51
	English	<u>−3.64</u>	−3.06	0.05	−0.89	0.36	1.21	−1.96
complex CoT	Chinese	0.39	2.86	−0.11	1.03	1.10	8.77	<u>−1.54</u>
	English	1.15	1.81	−1.01	3.25	0.41	1.20	<u>−1.46</u>
one shot	Chinese	−0.41	<u>−5.23</u>	−4.11	2.57	0.88	6.14	−0.85
	English	−0.93	−2.36	−3.03	0.88	2.47	6.23	<u>−4.42</u>

Note: Bold and underlined values represent the maximum and minimum values, respectively. Bold and underlined values represent the maximum and minimum values, respectively.
Abbreviations: Cl, claude-3.5-sonnet; Ds, deepseek-vl2; G4, gpt-4o-2024-11-20; Ge, gemini-2.0-flash; Gl, glm-4v-plus-0111; Hv, hunyuan-vision; Qw, qwen-vl-plus-0102.

varied preferences for prompt engineering strategies (Jullien et al. 2025; Lu et al. 2024; Nb and Nb 2023).

In summary, although complex CoT yielded moderate gains in selected tasks, their overall effectiveness in enhancing MLLMs' spatial reasoning ability remains limited. The persistent reliance on textual cues, difficulty in processing multimodal inputs, and lack of semantic integration in visually grounded contexts suggest that prompt engineering alone cannot overcome the fundamental perceptual and cognitive limitations of current models. Also, different training architectures induce distinct prompt-engineering preferences across models. These findings highlight the need for more robust architectures and training paradigms tailored to multimodal spatial reasoning.

5.4 | Comparative Analysis of GPT-5 and GPT-4o on GVSA

To examine advancements within the same model family, we compared GPT-4o with its successor, GPT-5, under both zero-shot and complex CoT conditions on the GVSA, evaluating performance separately in English and Chinese. As shown in Figure 15, in Chinese, GPT-5 achieves 48.44% compared with GPT-4o's 45.96% in the zero-shot, and 54.22% versus 44.49% with complex CoT, indicating a modest zero-shot improvement and a substantial complex CoT gain. A similar pattern is observed in English. These results demonstrate the incremental evolution of model architectures, leading to more reliable performance on complex and multi-step visuospatial tasks.

5.5 | Comparison on GeoAI Benchmark Datasets

Table 10 compares recent GeoAI benchmark datasets in terms of input modality, count, core tasks, and evaluation metrics. Existing benchmarks mainly adopt text-only inputs and target text-based geospatial skills such as geospatial Questions & Answers (Q&A) and code generation. In contrast, our benchmark introduces both text and image modalities, focusing on visuospatial abilities. The GVSABench covers 851 items that

jointly assess spatial abilities in both Geo and non-Geo scenarios and taps into more fundamental spatial capabilities. We also include consistency to capture the robustness and bilingual (Chinese and English) generalization of LLMs' spatial reasoning.

5.6 | Separation in Geo and Non-Geo Tasks

A Geo versus non-Geo separation phenomenon is observed in more than half of models, but with a reversal for Deepseek-VL2. As shown in Table 11, GPT-4o and GLM-4V scored 9.04% and 10.09% lower on non-Geo tasks than on Geo tasks, whereas Deepseek-VL2 gained 13.16% on non-Geo tasks. Hunyuan-vision and Qwen-VL displayed no clear scale-separation effect.

For GPT-4o, GLM-VL, Claude-3.5 and Gemini-2.0, the discrepancy is likely attributable to the training data of current MLLMs, which predominantly consist of natural images. Such data biases constrain the model's perception of small-scale abstract patterns (non-geo tasks), resulting in lower accuracy while conferring an advantage on geographical tasks that require interpreting complex satellite imagery and cartographic visualizations (Peng et al. 2021; Fang et al. 2022; Gao et al. 2025). This finding aligns with (Y. Zhang, Bai, et al. 2024), who reported that VLMs struggle with abstract patterns in Raven's Progressive Matrices (RPM) due to their reliance on natural images during training, reducing their sensitivity to non-naturalistic patterns.

By contrast, Deepseek-VL2 exhibited the opposite trend, achieving higher accuracy on non-Geo tasks than on Geo tasks. One plausible explanation is that its pre-training data may contain a larger proportion of synthetic and abstract images. Recent studies demonstrate that targeted exposure to synthetic and abstract visual data, such as symbolic diagrams, structured scenes, and schematics, can substantially improve VLM performance on abstract reasoning benchmarks and enhance non-naturalistic pattern recognition and spatial reasoning (Chen, Xu, et al. 2024; K. Zhu et al. 2025). Deepseek-VL2 may benefit from a similar data regimen, although its exact training corpus has not been fully publicly disclosed.

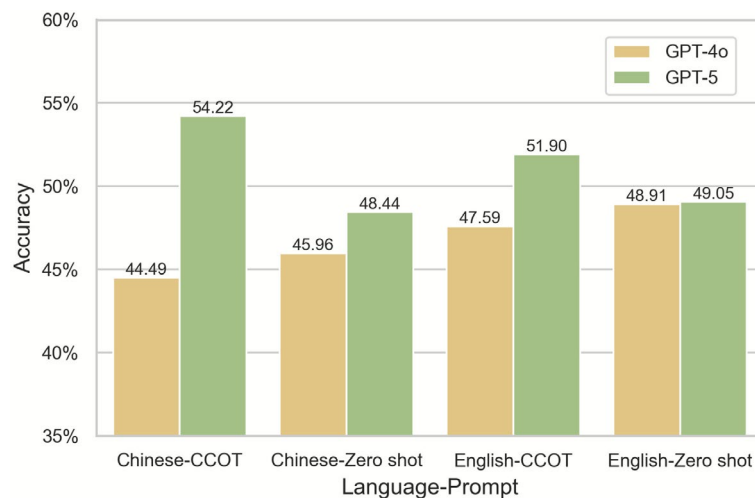


FIGURE 15 | Accuracy comparison of GPT-4o and GPT-5 under zero-shot and Complex COT.

TABLE 10 | Comparison of GeoAI benchmark datasets and evaluation metrics.

Dataset	Input modality	Item count	Language	Core tasks	Evaluation metrics
Hochmair et al. (2024)	Text	76	English	7-class spatial Q&A (literacy, GIS theory, coding, mapping, reasoning, etc.)	Correctness, consistency, word count of response
BB-GeoEval (Zhang, Wang, et al. 2024)	Text	750	English	Geospatial Q&A covering surveying, remote sensing, cartography, geodatabase, spatial analysis	Accuracy, GPT-4 judged quality, expert evaluation
STBench (Li, Yao, et al. 2024)	Text	13 distinct tasks and over 60,000 QA pairs	English	Spatio-temporal analysis capability: knowledge comprehension, spatio-temporal reasoning, accurate computation, and downstream applications	Accuracy, absolute error (meters)
Xu, Zhao, et al. (2025)	Text	900	English	12 geospatial skills: GIS concepts, NL2API, code explanation/generation/translation, route-plan, toponym, spatial understanding, etc.	Weighted accuracy
GeoAnalystBench (Zhang, Gao, et al. 2025)	Text	50	English	GIS workflow and code generation for spatial analysis; Python-based geoprocessing tasks	Validity, MAD, text similarity, codeBLEU
Yang, Fu, et al. (2025)	Text	105	English	Spatial cognition: Landmark/route/survey knowledge conversion	Accuracy, answer length
Ours	Text and image	851	Bilingual (Chinese and English)	Eight dimensions in spatial ability in both Geo and non-Geo tasks	Accuracy, consistency

5.7 | Performance on Artificial Versus Non-Artificial Tests

As noted in Section 3.2.2, we used the Seedream 4.0 (Seedream et al. 2025) to synthesize virtual images and maps, generating fictitious evaluation items to test the model robustness across real-world and virtual scenarios. Seedream 4.0 enables generation of diverse evaluation items via text-image fusion and multi-reference inputs. Its performance demonstrated a significant

lead over models like GPT-4o in the AGI-Eval text-to-image benchmark (Seedream et al. 2025). Using its web interface, we supplied textual descriptions to generate corresponding images. The resulting synthetic images fall into two categories: (1) street scene images (Figure 16a), incorporated into the intersection decision-making and spatial localization tasks, and (2) synthetic thematic maps (Figure 16b), used in the spatial deconstruction and spatial relation tasks. Each Geo subtype contained 20 artificial tests, yielding a total of 80 items.

TABLE 11 | The accuracy and accuracy difference (%) across all models on scale.

Model	Cl	Ge	G4	Gl	Hv	Qw	Ds
Non-Geo	43.04	50.84	43.50	38.37	38.28	<u>37.06</u>	49.34
Geo	45.45	52.20	52.54	49.27	38.07	<u>37.25</u>	36.18
Difference	−2.41	−1.36	−9.04	<u>−10.90</u>	0.21	−0.19	13.16

Note: Bold and underlined values represent the maximum and minimum values, respectively. Positive differences indicate higher accuracy in non-Geo tasks, while negative differences indicate higher accuracy in Geo tasks.
Abbreviations: Cl, claude-3.5-sonnet; Ds, deepseek-vl2; G4, gpt-4o-2024-11-20; Ge, gemini-2.0-flash; Gl, glm-4v-plus-0111; Hv, hunyuan-vision; Qw, qwen-vl-plus-0102.



FIGURE 16 | Example of synthetic street-view image (a) and thematic maps (b) generated by Seedream 4.0 (Seedream et al. 2025).

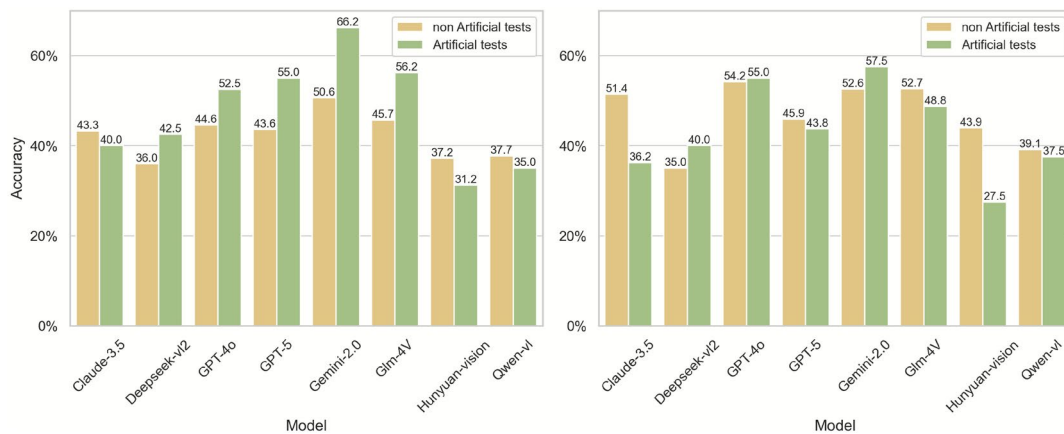


FIGURE 17 | Accuracy comparison on artificial tests and non-artificial tests under Chinese (left) and English (right).

We compared zero-shot accuracy across models on artificial versus non-artificial items within the Geo tasks (Figure 17). In Chinese, Deepseek, GLM, GPT-4o, GPT-5, and Gemini all scored higher on artificial tests, with gains of 6.5%, 10.5%, 7.9%, 11.4%, and 15.6%, respectively. In contrast, Hunyuan showed a 6.0% drop on artificial items. In English, only Claude and Hunyuan exhibited large deviations: their accuracy dropped by 15.2% and 16.4%, whereas the other models showed negligible differences.

Models that performed comparably across both artificial and non-artificial items demonstrate stronger generalization (e.g., Deepseek, GLM, GPT-4o, GPT-5, and Gemini). The stable performance of these models also demonstrates that synthetic images can serve as an effective tool for evaluating generalization beyond real-world datasets. Conversely, models showing large performance gaps (Hunyuan and Claude) appeared more sensitive to domain shifts. Their poorer performance on synthetic items may also reflect potential data contamination (Sainz et al. 2023; Cheng et al. 2025; Yao et al. 2024; Balloccu et al. 2024): models may have been exposed during training to real-world maps or street-view imagery similar to the non-artificial items, inflating their scores on those tasks while not benefiting from synthetic stimuli (C. Xu et al. 2024; Kocigit et al. 2025).

5.8 | Impacts of Input Language on the Model's Performance

As shown in Sections 4.1.1, comparisons between English and Chinese inputs across models reveal small but insignificant overall accuracy differences, indicating generally robust performance across input languages in GVSA tasks. However, the Geo tasks exhibit more pronounced variations, with 2.6% accuracy gaps favoring English prompts, whereas non-Geo tasks show negligible differences (mean=0.81%). These domain-specific patterns suggest heightened language sensitivity in geographic knowledge processing, potentially arising from imbalances in training data in which English geographic terminology is predominant (Li, Haider, and Callison-Burch 2024; Stillman and Kruspe 2025). This underscores the need for targeted multilingual fine-tuning strategies to reduce such biases and enhance model robustness for cross-lingual geographic applications.

5.9 | Limitations

This study assessed seven state-of-the-art MLLMs using a GVSA classification framework. Unlike prior work that focused only on geographic knowledge or visual-spatial skills in isolation, this dataset links both domains for a more integrated evaluation. However, it remains incomplete, as additional categories such as map planning (Ekstrom et al. 1976), trajectory prediction (Li, Haider, and Callison-Burch 2024), and scene understanding (Danish et al. 2025) were not included. The current framework used a binary Geo versus non-Geo distinction, which does not fully reflect the multi-scale and continuous nature of real-world geographic tasks and calls for more fine-grained categorizations in future work.

In addition, the dataset did not cover the full diversity of geographic data formats such as satellite imagery or digital elevation models, which constrains the generalizability of our findings across different data formats. Dataset scale and diversity constraints relative to real-world spatial reasoning scenarios remain a limitation. The GVSA benchmark currently lacks dynamic scenes (e.g., moving objects, temporal sequences, and interactive environments) that more accurately characterize practical GeoAI applications and real-world geospatial cognition. Potential cultural and language biases in geographic task formulations may also affect broader applicability. Finally, limitations in task-complexity coverage and cross-model generalization underscore opportunities for future expansion, as the current dataset did not include a systematic assessment of task difficulty.

Prompt engineering strategies (zero-shot, one-shot, and CoT) were tested but proved insufficient for fully revealing models' spatial abilities. Recent research suggests more advanced approaches (C. Li et al. 2025; Sprague et al. 2025), such as MVOT (C. Li et al. 2025), which visualizes reasoning traces, and SpatialCoT (Liu et al. 2025), which grounds reasoning in spatial coordinates. Future work should incorporate these methods to better evaluate and enhance MLLMs' spatial reasoning. Moreover, this study evaluates models primarily based on final answers but lacks a standardized protocol for assessing intermediate spatial reasoning processes, despite evidence that identical final responses can reflect varying reasoning quality.

Developing rigorous and standardized evaluation strategies remains an important direction for future work.

6 | Conclusion

This study introduced GVSABench, a comprehensive GVSA benchmark that integrates diverse categories of spatial tasks across dual scales, enabling a systematic evaluation of seven state-of-the-art multimodal large language models under zero-shot, zero-CoT, and one-shot prompting strategies. The results reveal several limitations: (1) overall accuracies remain moderate to low with substantial variation across models and languages; (2) cross-model consistency differs markedly; (3) prompt strategies provide little or no consistent improvement, indicating that engineering alone cannot resolve spatial cognition deficits; and (4) a clear Geo versus non-Geo (scale) separation effect emerges in both accuracy and prompt efficacy. Collectively, these findings highlight the nascent state of spatial ability in current foundation models and delineate critical gaps (cross-scale generalization, visual perception and deep spatial understanding) that must be addressed. By providing a systematic dataset and reproducible evaluation framework, this work offers new insights into the spatial-reasoning potential of existing models and paves the way toward more robust and cognitively grounded spatial intelligence.

Funding

This research is supported by the National Natural Science Foundation of China (NSFC; Grant Nos. 42371455, 42501551 and 42230103).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data and code that support the findings of this study are available at github: https://github.com/smileliaohua/GVSA_Benchmark.

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Appendix 1

The eight GSVA task categories are detailed below with examples:

Non-Geo Intrinsic-Static (NG-IS)

The NG-IS category evaluates models' ability to understand and process spatial information with non-Geo context. Tasks typically involve analyzing the spatial attributes and composition of individual or small objects, allowing for a comprehensive assessment of the model's performance in handling small-scale spatial relations. This category includes two types of test questions: Hidden Figures Test and Hidden Patterns Test (Ekstrom et al. 1976). Both tests require the model to infer and recognize simple patterns within complex ones. Example questions include:

Q: This is a test of your ability to tell which one of five simple figures can be found in a more complex pattern. At the top of each page in this

test are five simple figures lettered A, B, C, D, and E. Beneath each row of figures is a page of patterns. Choose the figure which you find in the complex pattern (Figure A1).

A: I. A; II. D.

Non-Geo Intrinsic-Dynamic (NG-ID)

The NG-ID category evaluates the model's dynamic processing of spatial information related to single objects. It examines the model's understanding of object properties and part-to-part relationships, while also considering temporal changes and spatial transformations such as movement, rotation, and folding. This category is broadly categorized into three types:

- a. Spatial rotation and perspective. Cube comparison, MRT (Peters et al. 1995), and the rotation and perspective tests in Purdue

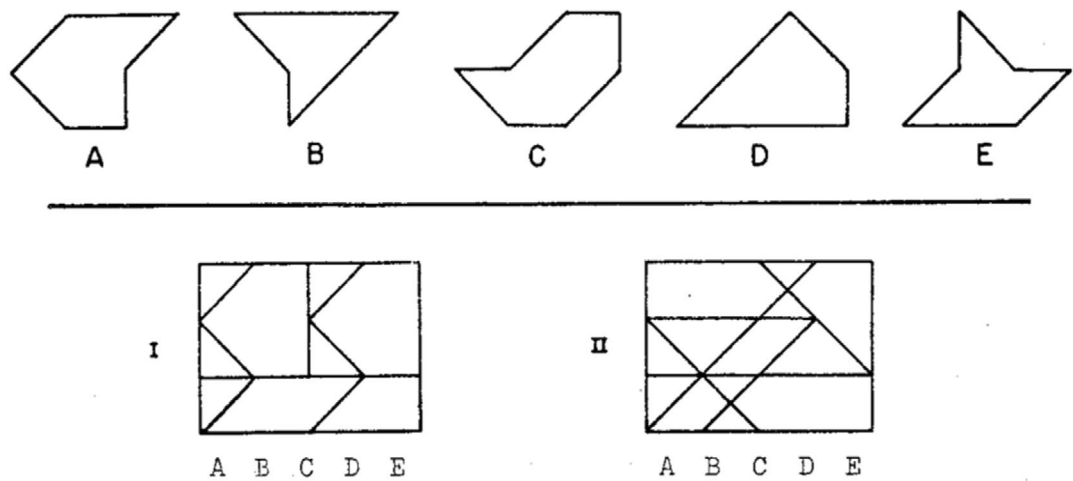


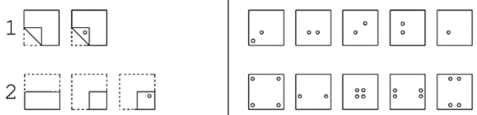
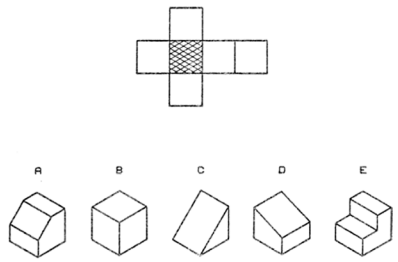
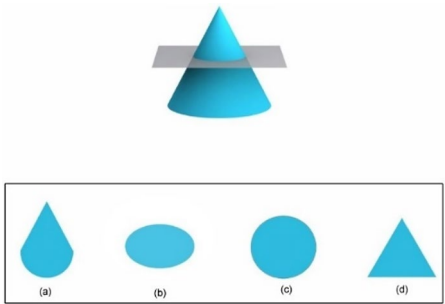
FIGURE A1 | Example of Hidden Figures Test.

TABLE A1 | Examples of NG-ID tasks.

Type	Description	Example
	Q: In the right 4 objects, find the exact match for the target object (far left) after rotation (there are 2 answers) A: 1. BC; 2. AD	
Spatial rotation and perspective	Q: Imagine yourself moving around the “glass box” until the black dot is located directly between you and the object. From this position, picture what the object looks like. Select from among the five drawings (A, B, C, D, or E) the one that matches the object as seen from the viewing position A: E	

(Continues)

TABLE A1 | (Continued)

Type	Description	Example
Spatial reasoning	Q: On the left is a sheet of paper folded and punched. On the right, find what it looks like when unfolded (A–E) A: 1. A; 2. D	
	Q: The diagram shows the inner surface of a three-dimensional object. The shaded area in the expanded drawing represents the bottom surface. Choose the object (A–E) that resembles a folded expansion A: B	
Spatial construction and deconstruction	Q: Select the cross-section visible when the gray cutting plane slices the object. Imagine facing the cutting plane head-on, as if looking in a mirror. Choose based on the shapes, not the sizes A: C	

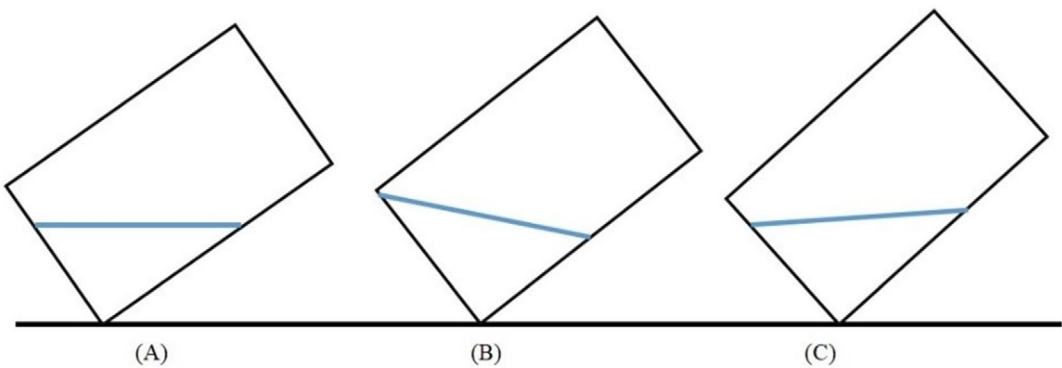


FIGURE A2 | An example of Water Level Test.

- Test (Bodner and Guay 1997) involve the ability to rotate two-dimensional (2D) and three-dimensional (3D) objects and identify their different perspectives, examining spatial visualization capabilities.
- Spatial reasoning. The Paper Folding Test (Ekstrom et al. 1976) requires the model to identify the position of holes in paper when folded and unfolded, testing the ability to understand spatial reasoning and dynamic changes.
 - Spatial construction and deconstruction. The CST (Cohen and Hegarty 2012) and The Development Test of Purdue Test (Bodner and Guay 1997) involves the transformation between 2D and 3D objects. These tasks require the model to construct a 3D object from a 2D template or deconstruct a 3D object into a 2D section, thereby assessing the model's spatial structure recognition and visualization capabilities. Examples of tests are shown in Table A1.

Non-Geo Extrinsic-Static (NG-ES)

The NG-ES category focuses on evaluating the model's ability to handle spatial relationships between objects and the environment, particularly static spatial information without geographic context. These tasks require the model to identify and understand the absolute position and layout of objects under environmental disturbances, while grasping abstract spatial principles such as horizontal invariance or verticality, without considering object movement or change. A typical task in this category is the Water Level Test (Lohaus et al. 1996), where the model must observe the water level in a tilted cup or a cup placed on an inclined plane and identify the horizontal line under these conditions. Examples include:

Q: As shown in Figure A2, select the most realistic horizontal line (A–C) in a glass of water at a certain tilt angle.

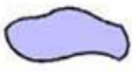
A: A.



FIGURE A3 | A question from PTSOT.

DIRECTIONS: Real world objects can be represented explicitly by point, line (arc), and area (polygon). Based on the examples below, classify the followings spatial data.

Example

Point 	Line  Arc 	Area 
Ex. trees, road intersections, poles in distribution networks.	Ex. roads, rivers.	Ex. the areal extent of a city, an area of a continent.

13. Locations of weather stations in Washington County _____ .
- (A) Lines
 - (B) Area
 - (C) Points and Lines
 - (D) Points and Area

FIGURE A4 | An example of comprehending geographical features.

Non-Geo Extrinsic-Dynamic (NG-ED)

The NG-ED category assesses the model's ability to handle relationships between objects and, particularly, spatial information at a small scale. It requires the model to visualize an environment from different

perspectives and account for changes in reference frames (Bednarz and Lee 2012). The model must recognize and understand the relative positions and layouts of objects while considering shifts in viewpoint and spatial references. A typical task is the PTSOT (Friedman et al. 2020), which requires the model to simulate a position at A,

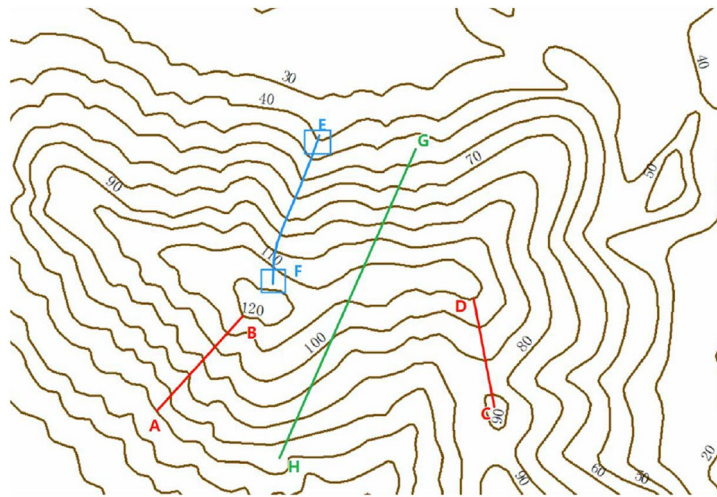
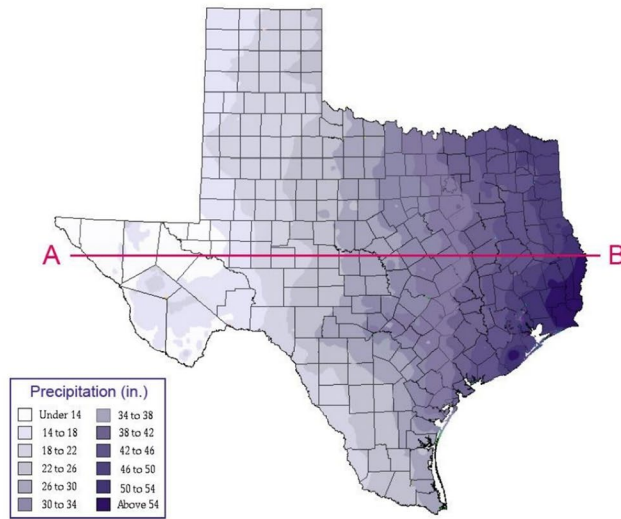


FIGURE A5 | Spatial construction.

Direction: The map below shows annual precipitation of Texas.



3. If you draw a graph showing change of Texas annual precipitation between A and B, the graph will be _____ .

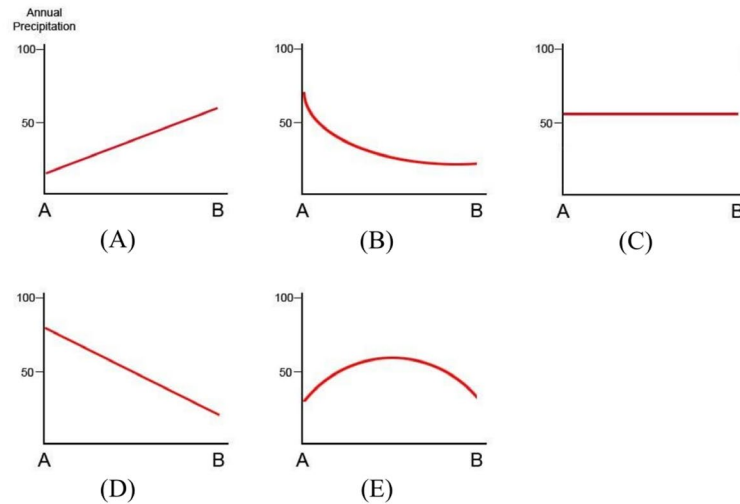


FIGURE A6 | Spatial deconstruction.

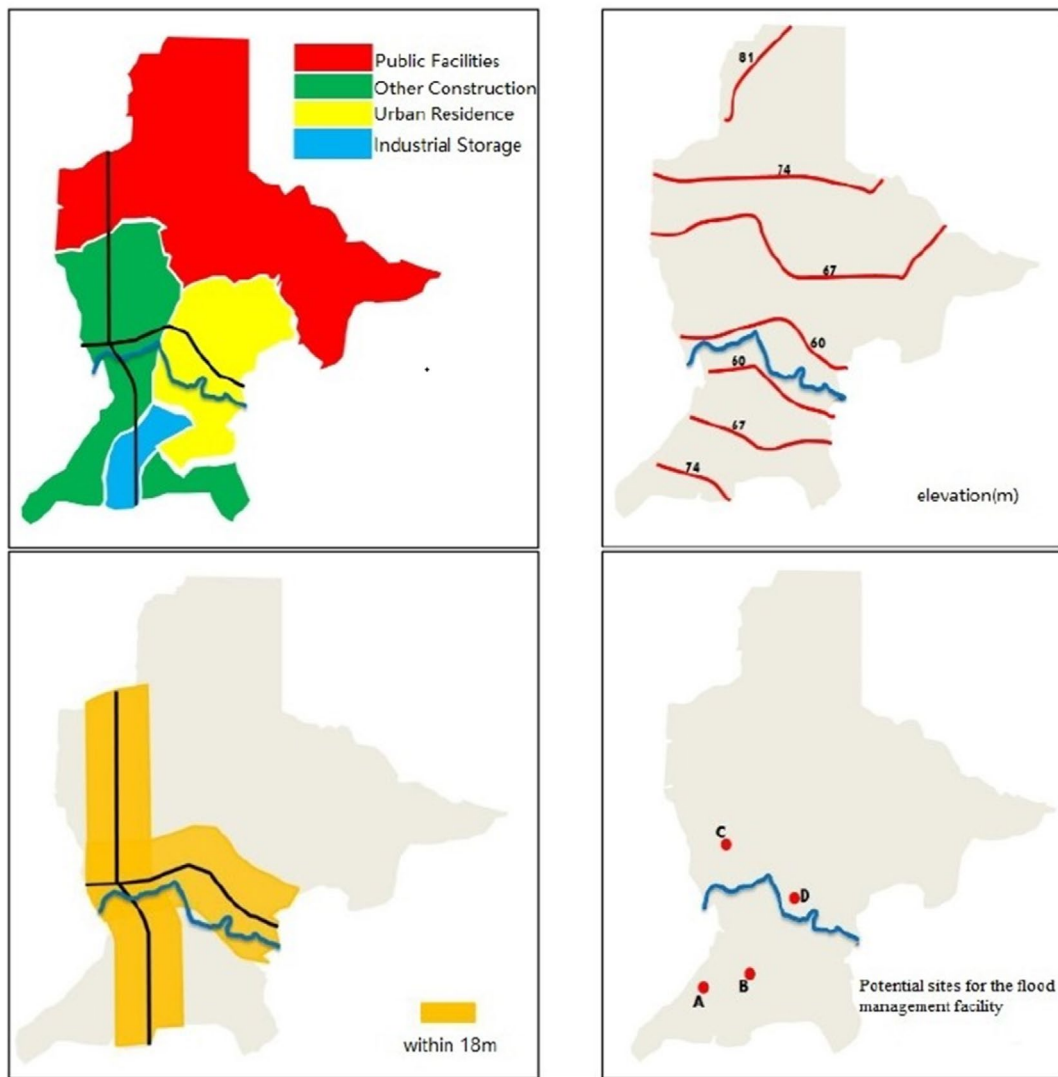


FIGURE A7 | An example of best location.

face toward B, and indicate an angle or direction from position C. Examples include:

Q: As shown in Figure A3, imagine you are standing in front of the car, facing the traffic light. Please indicate the stop sign's angle (clockwise 0° – 360°).

A: 123° .

Geo Intrinsic-Static (G-IS)

The G-IS category focuses on assessing the model's ability to process and understand geospatial information. Tasks in this category are crucial for improving the model's use in areas such as map reading, geographic information systems (GIS), environmental monitoring, and urban planning. Comprehending geographical features (Bednarz and Lee 2012) is a typical task, which requires the model to interpret geographical features based on spatial data, such as the location of weather stations, urban areas, and continental extents. This task not only examines the model's understanding of basic geographical elements but also its ability to accurately classify these elements based on their intrinsic properties. An example is shown in Figure A4.

Geo Intrinsic-Dynamic (G-ID)

The G-ID category evaluates the model's ability to process dynamic geospatial information. It requires the model to identify basic geospatial

elements (points, lines, and surfaces) and account for their changes, including spatial construction and deconstruction.

Spatial construction refers to the process of extracting the spatial attributes of geographic events from text descriptions or images and using illustrations to establish spatial structures or models. Oppositely, spatial deconstruction involves interpreting and refining spatial attributes from images or text. For example, the model may judge the direction and speed of water flow according to a contour map, reflecting spatial construction, or summarize rules of geographical phenomena using thematic maps, reflecting spatial deconstruction (Bednarz and Lee 2012; Newcombe et al. 2015). Examples include:

a. Spatial construction

Q: As shown in Figure A5, according to the topographic map, answer the questions: (1) which is steeper, AB or CD? (2) What direction is the water flowing between EF? Is the water flowing faster around F or around E? (3) is HG intervisible?

A: 1. AB; 2. F to E, at E; 3. HG is not intervisible.

b. Spatial deconstruction

Q: As shown in Figure A6, read the map of a geographical phenomenon on a global scale. Which of the following maps summarizes its distribution?

A: A.

DIRECTIONS: The following two maps show (A) Acres of corn production  (B) Value of hogs and pigs as percent of total market value of agricultural products sold.



7. If you draw a graph showing the relationship between map (A) and (B), the graph will be _____.

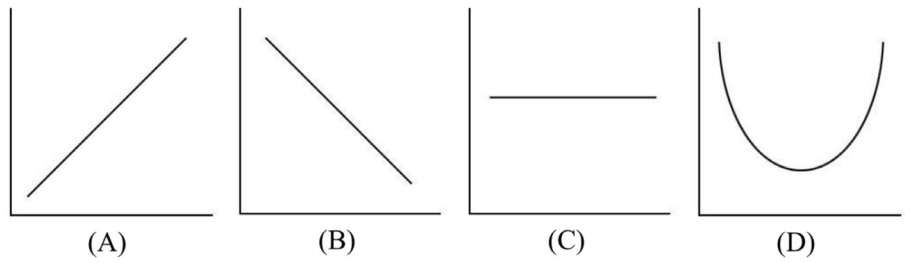


FIGURE A8 | An example of spatial relation.



FIGURE A9 | An example of map location.

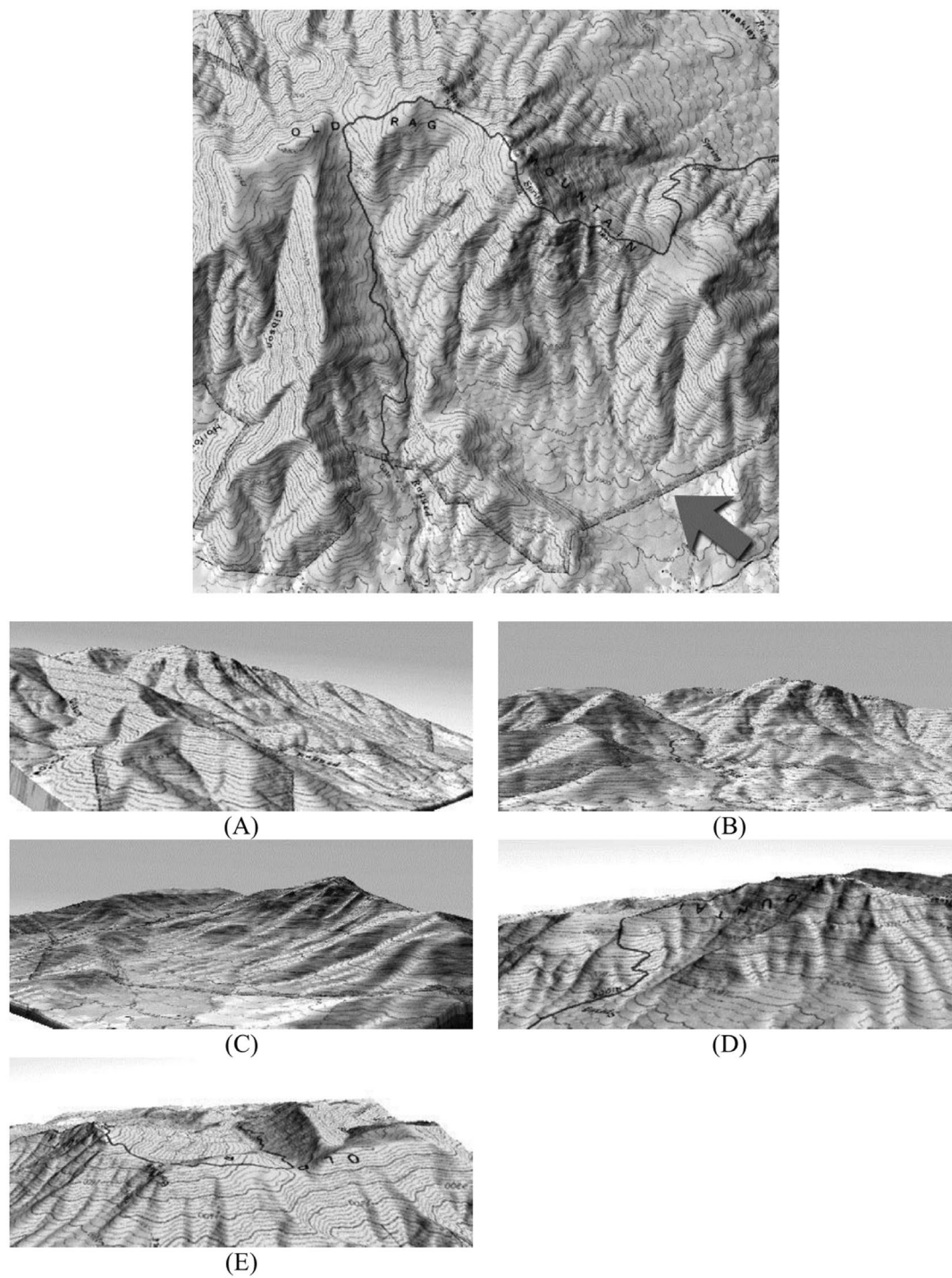


FIGURE A10 | An example of spatial positioning and orientation.

Geo Extrinsic-Static (G-ES)

The G-ES category focuses on assessing the ability of models to understand and analyze the relationships between objects and the environment at a large scale. These tasks require the model to recognize the relative positions and layouts of objects and infer their spatial relationships. The category includes three types of test questions: Best Location, Spatial Relation (Jongwon Lee and Bednarz 2009; Bednarz and Lee 2012), and Map Location.

- a. In the Best Location task, the model must evaluate the advantages and disadvantages of different geographical locations, considering various factors such as population distribution, transportation networks, and related facilities.

- b. In the Spatial Relation task, the model infers spatial relationships between map features, identifying and matching correlated regions or locations (e.g., finding the most similar area to a given image).
- c. The Map Location task requires the model to locate a real-world scene image on a map by recognizing and matching features from both the map and the scene, involving an understanding of the relative relationship between the features.

Examples of tasks are as follows:

Q: As shown in Figure A7. Find the best location for a flood management facility based on the following conditions. (1) The site should be within 60ft of an existing electric line; (2) The site should be located less

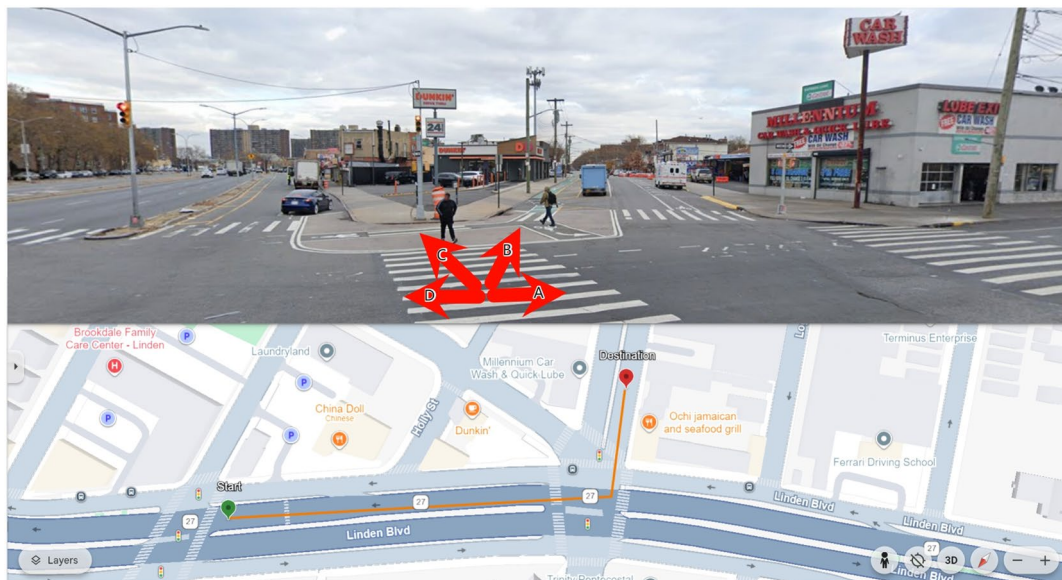


FIGURE A11 | An example of intersection decision-making.



FIGURE A12 | Examples of AI-generated street view images. (a) A correct image; (b) poor spatial alignment (McDonald's and Domino's incorrectly shown as adjacent).

than 220ft from the nearest road; and (3) The site should be located in State Park or Public Land. Choose the best site from options A-E for the flood management facility on the map.

A: C.

Q: As shown in Figure A8, As shown in Figure A8, find the map that has the strongest positive correlation with the left image (find the closest match).

A: D.

Q: As shown in Figure A9, select the location (A-D in red) that is closest to the real image on the map.

A: A.

Geo Extrinsic-Dynamic (G-ED)

The G-ED category assesses the model's ability to process dynamic changes in geospatial information. It requires the model to identify the relative positions and layouts of features while accounting for shifts in spatial perspective and reference systems. Key tasks in this category include Spatial Positioning and Orientation, Map Overlay (Bednarz and Lee 2012), and Intersection Decision-Making.

- a. In the Spatial Positioning and Orientation task, the model performs spatial reasoning based on geographic data and instructions to determine its position and direction.
- b. The Map Overlay task requires the model to integrate map information from multiple sources to provide a more accurate and comprehensive understanding of the geospatial data.
- c. In the Intersection Decision-Making task, the model makes correct turning choices at complex intersections based on the map, the given path, and a real-world scene image.

Examples of tasks are as follows:

Q: As shown in Figure A10, if you look at the area below in the direction of the arrow, which terrain view (A-E) most closely represents what you would see?

A: B.

Q: According to the route from the starting point to the destination on the map, choose the correct direction to reach the end point in the real scene map (Figure A11).

A: B.

Appendix 2

Below are the prompts of generating street images. Example outputs are shown in Figure A12.

Generate a daytime street view image with the following specific requirements:

1. Viewpoint: Use an eye-level perspective at human height, presented in first-person view.
2. Distance: Keep the camera close to the street to enhance immersion, as if the observer is walking into the street.
3. Environmental Features: The street view should include clear surrounding buildings, shops, and possible pedestrians or vehicles to showcase the street's vibrancy and characteristics.
4. Special Emphasis: The relative positions of each shop in the image must be accurate, with consistent distances and arrangement order between shops to allow observers to quickly identify the street layout through shop position relationships. Additionally, the relative positions of intersections should be accurately presented, including intersection directions, positions relative to surrounding shops, and traffic signs, enabling observers to quickly locate their position on the street.

5. Reference Location: The street view position should match the specific location on the provided reference map (attachment), with consistent roads to ensure geographic accuracy.
6. Details: The image should contain sufficient details for identifying specific street elements like shop signs and window displays, while ensuring these details accurately reflect the relative positional relationships of shops and intersections. Please ensure the generated street view image is consistent with the location and environmental features on the reference map, allowing observers to quickly recognize and locate themselves.